

Campus Navigation Game

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Campus Navigation Game

BSc (Hons) Computer Science

01/05/2026

Statement of Originality

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Abstract

This project aimed to design and implement a videogame based on the Lancaster University campus in order to induce improvement in navigational ability by encouraging exploration of a virtual representation of the Lancaster campus. The goal was to investigate whether a gamified approach to campus orientation could improve students' ability to navigate around campus using landmarks. An open-world, 3D first person game was developed, with the core game loop of a cooking and delivery pipeline. NPC and quests were introduced to further encourage exploration. A user study was then conducted to evaluate how well the videogame achieved its aims, by getting participants to do a navigation task before and after playing the videogame. Results from this study were inconclusive due to limitations with its design and sample size.

Contents

Statement of Originality.....	2
Abstract.....	3
1. Introduction	6
2. Background.....	8
3. Design.....	14
3.1. Game Genre	14
3.2. Art Style.....	16
3.3. Game Loop	18
3.4. Landmarks	19
3.5. Game Engine.....	26
3.6. Restaurant Minigame.....	26
3.6. Other Mechanics	36
3.7. Game Rules	39
3.8. Subsystem Design	40
3.9. Data Design.....	44
4. Implementation.....	45
4.1 Process	45
4.2. Testing	47
5. Evaluation.....	49
5.1. Design	49
5.2. Results.....	51
5.3. Discussion.....	54
6. Conclusions	57
6.1. Review of Aims	57
6.2. Revisions of Design and Implementation	57
6.3. Future Work	57
6.4. Personal Reflections	57
References	59
Appendix A. Project Proposal.....	62
Appendix B. Ethics Application.....	69
Appendix C. Ethics Form.....	73
Appendix D. Participant Information Sheet	74
Appendix E. Evaluation Walkthrough.....	77
Appendix F. Evaluation Questionnaire.....	80
.....	81

Appendix G. Evaluation Think Aloud Task Transcripts.....	88
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1. Introduction

The aim of this project is to develop and evaluate a videogame for use as a campus orientation technique, specifically to help students learn the campus layout of Lancaster University. Campus orientation is the period before the start of the academic year at a university or college. At Lancaster University, this comprises of a “Welcome Week” before the start of term, which allows students to “orientate ... on campus”, “settle into ... accommodation” and “get settled” in general (Lancaster University, 2026b). A central aim of this process is to help students learn the campus so that they can independently navigate to lectures, seminars and other events and locations.

Existing campus orientation methods such as the “Welcome Week”, which includes in-person campus tours, have their limitations. Shupp (2009) suggests that traditional campus orientation methods need to be replaced by a “comprehensive process”, reaching students on an “on-going basis”, rather than just for a single week. CampusMind (2025) suggests that traditional, week-long orientation methods are ineffective, as information delivered in short timeframes is unlikely to be retained long term – known as information overload. Ebbinghaus’s forgetting curve, successfully replicated by Murre and Dross (2015) demonstrates that humans forget 90% of new information within the first seven days. When applied to campus orientation, this suggests that students may struggle to retain spatial information delivered during a single campus tour or during the “Welcome Week”. As Shupp suggests, a more “comprehensive process” is required.

Other techniques have been explored, such as virtual campus tours, both using Virtual Reality (VR), an approach explored by Suwarno and Munarka (2020), and with a simple 3D program, an approach explored by Larmore et al (2005). These solutions do allow students to familiarise themselves with the campus before the start of term, reducing information overload. However, these approaches lack reusability, sustained engagement and motivation for extended student use. In contrast, videogames are designed around replay-ability, and have been found to have some associations with navigational ability (Coutrot *et al.*, 2019)(Murias *et al.*, 2016). Therefore, gamification could be used to harness these benefits. Defined by Deterding et al (2011) as “the use of design elements characteristic for games in non-game contexts”, gamification has shown positive signs, especially on engagement and enjoyment (Hamari, Koivisto and Sarsa, 2014).

A gamified campus orientation solution has the potential for several benefits. Firstly, it may improve navigational ability through repeated exploration of a virtual campus. Secondly, it may reduce information overload by distributing campus learning over multiple play sessions. If successful, the findings of this project may inform further videogames designed for other universities, and in general inform gamified solutions to campus orientation.

The aims of this project are to:

- Design and implement a videogame modelled on the Lancaster University campus.
- Incorporate gameplay intended to promote repeated exploration of the virtual campus and repeated exposure to campus landmarks.
- Conduct a user study to measure the impact of the videogame on navigational ability of the campus.

The structure of the project is as follows. Chapter 2 will contain the background. This section will discuss the background research on campus orientation, videogames for navigation and

gamification. It will analyse the research for any gaps and any implications for this project. Research from this section suggests that existing campus orientation techniques are limited. Videogames and gamification are useful for learning, and for navigation. Therefore, a gamified system for campus navigation is a valid solution for the lack of comprehensive campus orientation techniques currently available. Chapter 3 will contain the design. This section will discuss key design considerations for the videogame. These include the game design, map design, art style and the software design. After consideration of the aims of the project, based on the background research, an open-world design was chosen, with a core cooking and delivery game loop, as well as NPC driven quests. A low poly art style was chosen, and key landmarks chosen to be modelled in more detail. Chapter 4 contains the implementation. This section will discuss how the design was implemented into a working system. It details how the subsystems from the design were well implemented, and despite encountering some bugs, the system was implemented as per the design. This section will also contain testing of the system, which found that the system largely met the requirement of running at 60 FPS on a normal PC, despite struggling slightly under heavy load. Chapter 5 will cover the design, results and discussion of a user study. The evaluation tested participants on a navigation task before and after playing the videogame, but found inconclusive results due to limitations with the study. Chapter 6 will contain the conclusion. This section will be a reflection on the aims of the project to see if they have been achieved. It will also discuss possibilities for future work building from this project, and finally some personal reflections. It found that the videogame needs a more realistic map and more focused gameplay to achieve its aims, but lessons were learnt.

2. Background

The background research on this topic will be presented in two subsections. Firstly, we will discuss the existing literature on campus orientation, and some proposed solutions to the problem. Secondly, we will discuss videogames for learning and how a videogame could be applied to the problem of campus orientation.

As mentioned previously, typical campus orientation involves a week-long “Welcome Week” which introduces students to the campus and often involves several mandatory and optional events. According to Shupp (2009), these orientation programs make a “substantial impact” on welcoming students to university, acting as a “bridge” to the “first stages of retention”. However, despite their impact, traditional campus orientation programs are not perfect. Shupp argues that “orientation must be considered a comprehensive process rather than a single event”. At Lancaster University, campus orientation is a week-long event. This approach is not sufficient. CampusMind (2025) cites information overload as one of the key reasons for this. According to Gross (1964), information overload “occurs when the amount of input to a system exceeds its processing capacity”. Traditional campus orientation presents a lot of information to students over a short period of time (days or even hours in some cases). According to Ebbinghaus’ Forgetting Curve, which has been successfully replicated by Murre and Dros (2015) and shown in Figure 2.1, humans forget an average of 90% of new information within the first seven days without reinforcement. Although students will be repeatedly visiting the campus, providing a level of reinforcement, students will have many other things to focus on, leading to the information overload effect. This supports the idea that campus orientation needs to be a comprehensive, continued process with opportunities for repetition and reinforcement.

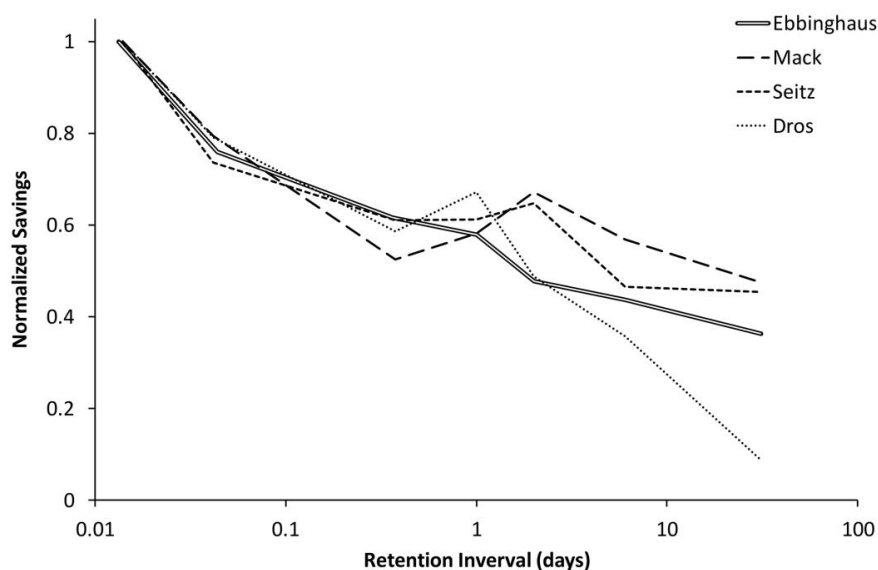


Figure 2.1 - Normalized savings scores as a function of retention interval on a logarithmic scale, rescaled so the first data point is 1.0 for all curves – comparing Ebbinghaus’ original Forgetting Curve to recreations by Mack, Seitz and Dros. (Murre and Dros, 2015)

First, we will discuss general campus orientation techniques that have been developed, not specific to campus navigation.

Several methods have been proposed to tackle the aforementioned requirements for a modern campus orientation system. One of these is the use of virtual reality (VR) to present general

faculty information, which saw an improvement in student knowledge retention due to the nature of video attention spans (Valenti, Lund and Wang, 2020). This study consisted of a test whereby elements of library service and library education were presented to one set of participants via an interactive VR experience, and other set of participants through a static website. The study found that VR to have a positive effect on students' interest. The Google Cardboard headsets used were inexpensive, so this method could be used as a continued orientation process. Another more popular method is an online student orientation (OSO). While OSOs are often designed for the orientation of online students, and have found success in that domain (Cho, 2012), they are also starting to be integrated into the orientation of in-person students. Although both of these methods allow for general reinforcement and continued contact with students, neither of them explore the key facet of campus orientation – learning the campus layout. As previously mentioned, traditional campus orientation methods rely on in person campus tours in order to help students learn the campus. Due to the aforementioned problems around information overload, students may not retain all of the campus information given in the campus tours. Therefore, students must rely on additional navigation tools either provided by the university or publicly available. At Lancaster University, students are provided with use of MazeMap (2026) to help them navigate the campus. This is a campus map with live GPS position information as well as internal and external building maps. At Lancaster University, MazeMap is built into the ILancaster mobile phone app that students use to check into lectures, shown in Figure 2.2. Use of GPS may be useful in the short term, but Dahmani and Bohbot (2020) have suggested that habitual use of GPS negatively impacts navigational ability, as it removes the requirements to use our internal navigation system and pay attention to our surroundings. The study assessed lifetime GPS experience of 50 regular drivers and found that the greater use of GPS, the greater decline in spatial memory over time. This suggests that relying on GPS navigation maps such as MazeMap is detrimental for the campus navigation ability of students. Therefore, a more comprehensive campus orientation technique focused on campus navigation is required.

Campus Navigation Game

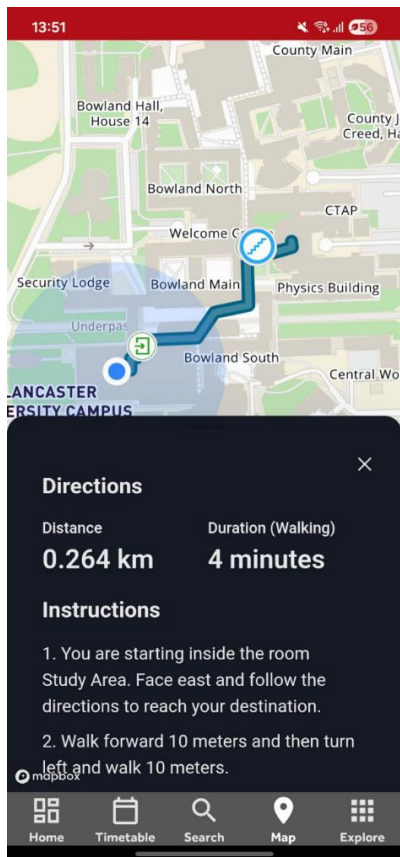


Figure 2.2 - An example use of MazeMap with the ILancaster app at Lancaster University, getting directions from the Library to the Faraday Complex

There has been some limited research into campus orientation techniques developed to help with campus navigation. The traditional method is in-person campus tours during the ‘Welcome Week’. These have limited effectiveness due to the aforementioned issues with information overload. Larmore et al (2005) presented a possible solution with their virtual campus program, shown in Figure 2.3. This consisted of a virtual 3D representation of the University of Nevada which allowed the user to explore the campus and go on virtual campus tours. The virtual campus could be presented to students before the start of term, allowing them to familiarize themselves with the campus before then, reducing disorientation in the ‘Welcome Week’.

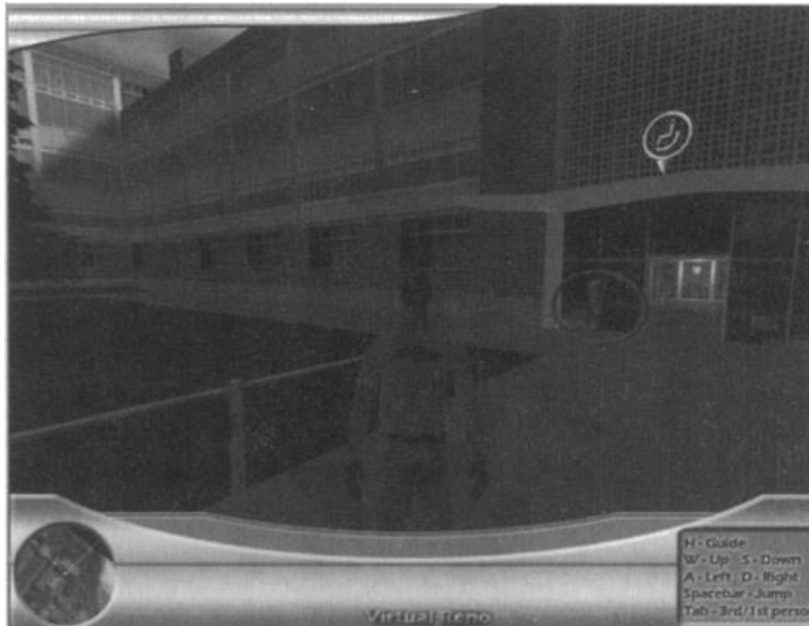


Figure 2.3 - A screenshot from Larmore et al's virtual campus program showing a player exploring the virtual campus on foot (Larmore et al., 2005)

Suwarno and Murnaka (2020) extended this virtual campus area of research by incorporating a VR approach, using real images from campus, shown in Figure 2.4. Information about the campus was presented using narration and visuals. Both of these solutions allow students to familiarise themselves with campus landmarks before the start of term. However, neither approach present much, if any, incentive to continue using the virtual campus programs. This is key, as repetition is key for memory and learning (Hintzman, 1976). If the student does not have the motivation to repeatedly use one of these virtual campus tools, then the effects will be limited. As CampusMind (2025) suggests, this is why the gamification of campus orientation can be useful.



Figure 2.4 - A screenshot from Suwarno et al's VR virtual campus program (Suwarno and Murnaka, 2020)

Research dating back to the 1980s has demonstrated the positive effects of videogames on traits such as reaction time, and research has found that they generally have a positive effect on learning (Kenwright, 2017). Research into the effect of videogames on navigational ability is ongoing and limited, but the research that does exist shows some positive signs. Murias et al (2016) found that videogame experience correlated with performance on a virtual navigation task. The participants who played videogames with a navigation aspect used more efficient navigation strategies. These findings are backed up by Coutrot et al (2019), who found that virtual navigational ability was predictive of real-world navigation performance. A similar study was conducted by Yavuz et al (2024), who also found that videogame experience was associated with wayfinding performance. This body of research tentatively suggests that videogames are a legitimate way to train navigational wayfinding ability. Furthermore, gamification in general has seen some positive effects, especially on engagement and enjoyment (Hamari, Koivisto and Sarsa, 2014), although not conclusively, as the “effects are greatly dependent on the context”. Gamification is defined by Deterding et al (2011) as “the use of design elements characteristic for games in non-game contexts”.

Based on the positive signs from research on videogames for navigation and gamification, a gamified solution to campus orientation could meet the criteria for a comprehensive, continued orientation process. Azis (2025) has presented a gamified solution for campus orientation, focusing on helping students learn the campus. Azis builds upon the virtual campus programs mentioned previously, but adds a gamified aspect, using Roblox to allow students to explore the virtual campus while interacting with non-playable characters (NPCs) who give missions to prompt further exploration. A screenshot from this videogame is shown in Figure 2.5. The videogame was found to “improve users’ spatial understanding of the campus”. A similar videogame developed for Lancaster University could yield similar benefits for students. Azis focuses the videogame around landmarks, which are “accepted as one of the vital elements in both virtual and real environments during wayfinding tasks” (Yesiltepe, Conroy Dalton and Ozbil Torun, 2021)



Figure 2.5 - A screenshot from Azis' Roblox campus navigation game, showing a campus building and a quest objective (AZIS, 2025)

This videogame builds upon Azis' system by taking inspiration from the concept of implicit learning. Azis presents a videogame which is clearly designed to teach the user about the campus layout. However, this presents a possible issue with student engagement, as students may not want to play the videogame if it is still heavily focused on being a virtual campus. A further layer of gamification could be added, gamifying the virtual campus so that although students will know it represents the campus, it takes on a different identity. This involves designing the virtual campus so that it resembles the original campus, but is not explicitly labelled as a university campus. Instead, it may be labelled as a village or a town. As previously mentioned, adding this layer of abstraction relies on implicit learning. Reber (1989) defines implicit learning as knowledge "acquired largely independently of the subject's awareness of either the process of acquisition or knowledge base ultimately acquired". We can align this definition with the proposed videogame, whereby the "process of acquisition" is the videogame, and the "knowledge base" is the spatial knowledge and mental map of Lancaster University campus. By shifting the focus from the videogame's underlying educational purpose, students may be more motivated to play the videogame. According to the principle of implicit learning, students will still gain the spatial knowledge of the campus even if they are not fully focused on that objective. Further research on implicit learning specifically for navigation discusses the human "ability to adapt to environmental constraints – to learn – in the absence of any knowledge about how the adaptation is achieved". This further suggests that the players of the videogame do not need to be focused on the objective of learning the campus environment, rather this can happen almost subconsciously whilst they play the videogame. Students will be aware that the map is modelled on the campus, but this does not need to be focused on or repeatedly stressed. This means that the videogame does not need to repeatedly explicitly label the map as a campus. If the videogame map matches the campus map, and contains the same landmarks, then players should gain spatial memory for this same map, whether that be virtually or in-person.

3. Design

The proposed videogame shall now be referred to as the “CNG”, meaning “Campus Navigation Game”.

The following are requirements developed from the background research and the aims of the project.

Functional requirements

1. The CNG shall contain a 3D map modelled on Lancaster University campus.
2. The map shall contain recognisable landmarks from the Lancaster University campus.
3. The CNG shall allow the player to control a character that can walk around the map.
4. The CNG shall include mechanisms that encourage the player to explore as much of the map as possible.
5. The CNG shall not focus on the map being a campus.

Non-functional requirements

1. The CNG should run at 60 fps minimum on a desktop PC with 16GB RAM, 6-core processor and a mid-range dedicated GPU at 1080p.
2. The CNG should be not cause confusion when used.
3. The CNG should be replay-able.

3.1. Game Genre

These requirements were formulated from the findings from background research. There were many possible approaches to meeting these requirements. The first stage of the design was to decide on the overarching game genre. A linear level-based game structure, where the player moves through discrete levels in turn, was a possible option. This genre provides strong control over where the player visits, and could ensure that all key locations on the campus are visited. However, it mostly prevents free exploration, and segments the map, which makes it more difficult for the player to form a spatial map of the entire campus. Instead, an open-world design was chosen, as it facilitates the requirements set out above. Open-world design usually centres around one large continuous map rather than separate levels. This matches the design of Lancaster University’s Bailrigg campus, which is one entity, as opposed to other campuses which are split over many locations. Modelling the campus in this way would allow for continuous navigation of the space, supporting spatial memory of the campus layout as a whole. With the entire game being set on one map, the spatial memory gain for that environment will be maximised, as opposed to having multiple levels. The largely unconstrained movement and navigation of players in open-world games also allows landmarks to be included and have meaningful navigational impact. Furthermore, allowing a level of unconstrained player movement means that there is no upper limit to the amount of exploration that a player can do across the map, allowing for maximum spatial memory gain.

With open-world chosen as the CNG game genre, the next step of design was to explore the plethora of existing videogames under the open-world genre that excel at encouraging repeated exploration of their game world, in order to gain some inspiration for further design of the system.

The Legend of Zelda: Breath of the Wild (Nintendo, 2017) was one of the first games that were explored as inspiration for the CNG. This game is an open-world game that encourages exploration and allows players to complete the story in a non-linear fashion. A screenshot is shown in Figure 2.1. This game meets the requirement of encouraging exploration, but does it

in an unstructured manner. This means that players explore the game map largely unguided. This method could be used in the CNG, however leaving exploration down to the player may mean that players do not explore the virtual campus fully, or do not explore the routes that students would be taking in real life. A more structured approach was therefore sought.



Figure 3.1 - Screenshot showing the open-world map in The Legend of Zelda: Breath of the Wild (Nintendo, 2017)

Stardew Valley (ConcernedApe, 2016) also came up as a possible inspiration. Stardew Valley is a farm life simulation game in which players inherit a farm and have to manage and grow the farm, whilst embarking on quests and interacting with villagers, shown in Figure 2.2. This videogame stood out as it is an open-ended game, allowing for unlimited replay-ability, which is one of the key requirements for this CNG. Furthermore, Stardew Valley encourages players to explore the map through the quest system, which is another key requirement. The farm life simulation aspect also fit the criteria, as it allowed for lots of progression and reason to continue playing the game. Stardew Valley also contains many recognisable landmarks across the map, which allows the player to develop spatial memory with continued playtime. The exploration in this game is also partly unstructured, with players expected to explore the map largely unguided. However, the use of NPC driven quests adds some structure, as many of these quests require players to navigate to certain areas of the map. Stardew Valley style quests therefore stood out as a possible game feature to encourage exploration.



Figure 3.2 - Screenshot of a player's farm in *Stardew Valley* (ConcernedApe, 2016)

Stardew Valley, with its partly structured exploration and farm life simulation game loop, was therefore cemented as the core inspiration for the CNG. This game provided a template for encouraging repeated exploration of the map, and provided some structure with the quest system.

3.2. Art Style

Taking inspiration from *Stardew Valley*, it was settled that the CNG should contain some sort of simulation game loop, along with an open-world game map and an NPC-driven quest system. Initially, *Stardew Valley* was also used as an inspiration for the art style of the CNG, with a 2D top down pixel art style. However, this does not meet the requirement for a 3D virtual campus map, and would therefore not match the perspective of a prospective student. Furthermore, using a 2D perspective means that variations in terrain cannot be modelled. The Lancaster University campus has recognisable variations in terrain, most importantly the large hill leading up to the main campus. Therefore, it was key for the game to be able to model terrain. Also, although *Stardew Valley* is an open-world game, it only displays part of the map at once, rather than one continuous map. This means that the map behaves more like a map split into different levels. As discussed previously, this is less effective at inducing spatial memory of the campus map as a whole. A HD2D style was explored as an alternative, taking inspiration from *Octopath Traveller* (Square Enix, 2018), which is a role-playing game with a similar pixel art style to *Stardew Valley*, but is modelled in a different perspective, shown in Figure 3.3. The HD2D perspective differs from the top down 2D perspective shown in *Stardew Valley* as it is in a fully 3D environment, allowing for terrain and other 3D effects, but uses billboard sprites and pixel art to create a 2D effect. Using HD2D would allow for the recognisable terrain of Lancaster University campus to be modelled, improving upon the top down 2D perspective and meeting the requirement of having a 3D campus map environment. This would be a continuous map, rather than being split into multiple portions like *Stardew Valley*. Initial prototypes of the system were modelled with this perspective and pixel art style.



Figure 3.3 - Screenshot from Octopath Traveler (Square Enix, 2018)

The pixel art style was initially chosen due to the inspiration taken from Stardew Valley and Octopath Traveller for other game elements. As these games both utilise the pixel art style, it seemed natural for the CNG to also use the pixel art style.

Experimentation with the HD2D perspective was unsatisfactory. Although this perspective allows for a 3D map to be modelled in the game, it does not match the real-life perspective that you would have when navigating the campus. Humans see in a first person view, rather than a 3rd person HD2D perspective. Furthermore, humans can rotate their head and bodies and look in 360 degrees, whereas a HD2D perspective is fixed at one position in 3rd person. Therefore, although the campus environment can be properly modelled in HD2D, the spatial memory learnt by playing the game would not be very similar to the spatial memory needed for navigation of the real world campus, due to these differing perspectives. To solve this difference, the perspective was changed from HD2D to a first person 3D perspective, to closer model the perspective of a real-world student navigating the campus.

Deciding to use a 3D perspective rather than a HD2D perspective also meant that the pixel art style of Stardew Valley and Octopath Traveller was not as applicable. A pixel art style is more suited to 2D or HD2D games, rather than 3D, due to its 2D, resolution dependent nature. Using pixel art in 3D would lead to inconsistent detail and a lack of immersion, due to the changing perspective and distance to objects when moving in 3D, and the fact that 2D pixel art does not scale in 3D. Therefore, a different art style was required to fit the new 3D perspective. There were many possibilities when choosing a new art style, such as realistic, low poly, cartoon, hand painted or retro. A low poly art style, similar to that of games such as Superhot (Superhot Team, 2016) - shown in Figure 3.4 - was chosen as the art style due for a variety of reasons. Firstly, low poly offers strong performance and is less taxing on the GPU, due to the low number of polygons used in the models, leading to faster rendering. This means that the game is more likely to meet the requirement of running at 30fps on a desktop PC, compared to art styles such as realistic, which use far more polygons and more sophisticated lighting. In addition, there are numerous non-copyright low poly models available on platforms such as Sketchfab, meaning that development time can be reduced and more time concentrated on

meeting the key requirements of the CNG. A realistic art style would have allowed for the most realistic representation of the campus, however requires long term modelling experience to create satisfactory game assets, experience not present in the developer. Low poly was therefore the best alternative to create memorable, recognisable representations of campus landmarks.



Figure 3.4 - A screenshot from Superhot (Superhot Team, 2016)

3.3. Game Loop

At this stage of the design, decisions had been made to include a low poly art style, first person 3D perspective and an open world loosely modelled on Stardew Valley's life simulation and quest system. The next part of design was to decide upon the key game loop. In Stardew Valley, the game loop is the player's farm. The player continually expands and upgrades the farm whilst exploring the map and completing quests. A similar game loop was explored, in order to make the game extremely replay-able. Initial ideas were to have the player play as a student with a dormitory that they could customise. They would have to complete a plethora of daily tasks such as cooking, laundry and attending lectures. Similar to upgrading your farm in Stardew Valley, the player could upgrade their room with new furniture, posters, and other items. This would provide an aspect of replay-ability, and would closely model the real-world navigation around campus. However, this idea was discarded as one of the key requirements of the CNG was that the game should not focus on the map being a campus, in order to provide better student engagement. Another game loop was that of the player owning a restaurant in a town. This would provide replay-ability similar to the farm simulation loop of Stardew Valley. This would consist of the player owning a restaurant whereby they could unlock new dishes, decoration and furniture, similar to the successful game loop in Stardew Valley where the player unlocks new crops, tools and decorations. Furthermore, a restaurant as the core game loop would allow for the addition of a delivery service, where the player not only has to cook meals, but also has to deliver them across the map. A delivery service adds another structured exploration mechanic, hitting the requirement of encouraging exploration across the map. Along with a quest system tied to NPCs, this would provide a sufficient amount of structured exploration, along with the unstructured exploration that comes with the open-world genre. A restaurant system was also more fitting for the map as much of the campus is paved. The player could therefore own a restaurant in the centre of the map, corresponding to Alexandra Square,

which is the focal point of the Lancaster University campus. Having players based in the centre of the map means that they should gain a good spatial map of all of the map, rather than being based in one corner of the map. After this consideration, a restaurant was chosen as the core game loop in the game.

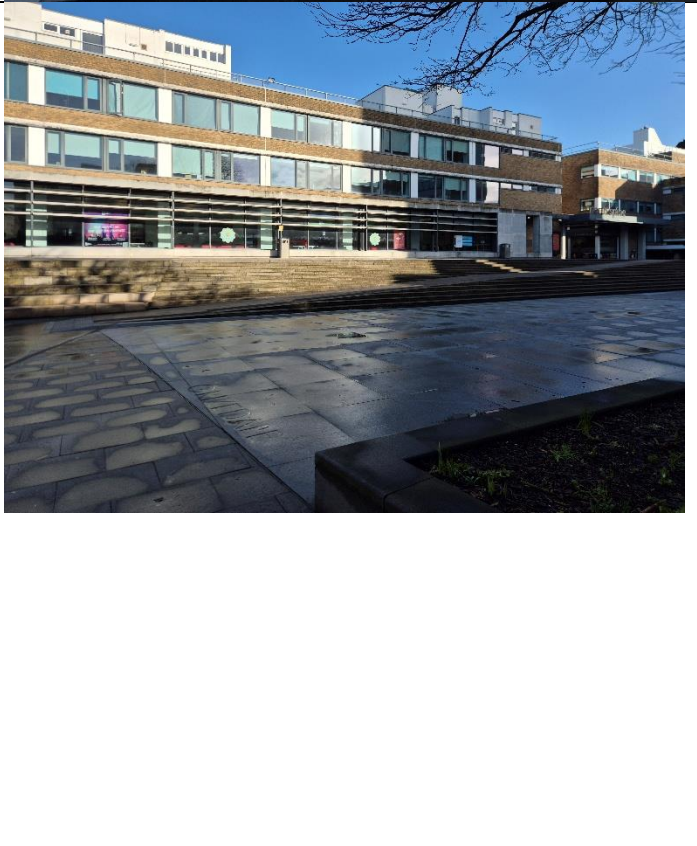
3.4. Landmarks

With the game loop decided, the decisions on the map had to be made. As mentioned in the requirements, the map had to be 3D and modelled on the Lancaster University campus. It should also contain recognisable landmarks from the campus. The initial plan was to model each building on the map in the low poly style to match the campus. This would have been the best solution as it would mean that the in game map would represent the real-world campus as closely as possible within the chosen art style. However, modelling each building on campus was too lengthy a process to fit within the time constraints of the project. As a compromise, landmarks were chosen in order fulfil that requirement of the CNG. The landmarks chosen, and the reasoning, are detailed in Table 3.1.

Table 3.1 – Chosen landmarks with justification

Landmark Chosen	Real-world Image	Reason For Inclusion
InfoLab21		Key landmark in South Campus, with a distinctive shape and recognisable blue and orange colour. Used by Computer Science students.
LICA		Key landmark in North Campus, with a recognisable green colour and large structure. Used by contemporary arts students.

SportLancaster		Key landmark for West Campus. Recognisable black colour and large shape. Used by people using the gym, swimming pool and those using squash, badminton and basketball courts. Used by many sports societies.
Sport Courts		Key landmark in West Campus. Used for basketball, netball and tennis. Seen when travelling in and out of the main road to campus.

Bowland Tower		Key landmark in Alexandra Square in the centre of campus. Tallest building on campus, used as accommodation.
Alexandra Square		Key landmark in the centre of campus. As well as containing the other landmarks of the library and Bowland Tower, it is also where many students gather on the steps. It also provides a central hub for shops and restaurants, and connected the two halves of campus through the North and South Spine.

Campus Navigation Game

Furness Accommodation		Distinctive landmark in South East campus due to the recognisable red colour of the accommodation blocks.
Fylde Accommodation		Distinctive landmark in South East campus due to the recognisable blue colour of the accommodation blocks.

Underpass		Key landmark underneath Alexandra Square, where buses to and from campus pass through.
Football and Rugby Fields		Key landmark of West Campus, with the fields running alongside the A6 on the west side of campus.
Chaplaincy		Key landmark of North Campus, with a distinctive architecture and white colour, used by religious groups and for music events.

Pendle-Grizedale Walkway		Key landmark for South Campus, with white pillars and building overhang linking Grizedale bar, co-op and the Pendle porters. Part of the South Spine, a popular route, and used by Grizedale and Pendle students regularly, as well as co-op shoppers.
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The chosen landmarks were to be modelled in more detail. The rest of the buildings on campus would be modelled, but not to the same degree of detail. This saves a lot of time whilst still creating a recognisable representation of the campus. Other buildings would be imported from the satellite images of the campus, in order to preserve the correct layout and distances and save time compared to modelling each building by hand. However, importing the campus layout from satellite data provides no terrain data. Terrain needed to be modelled in order to make the virtual representation of campus more recognisable. Lancaster University's campus has a large hill (Bigforth Drive) running up onto the main campus, shown in Figure 3.5. Modelling this, and other variations in terrain, was necessary to ensure proper spatial memory transfer between the videogame map and the real-world campus. A further required addition was that of trees and foliage. The Lancaster University campus is shrouded by trees, and in general prides itself in being green, having been awarded the Green Flag award for quality green spaces (Lancaster University, 2026a) This therefore needed to be reflected in the game with the addition of trees and foliage. Other key environmental features on the campus, such as Lake Carter, shown in Figure 3.6, also needed to be modelled.

Campus Navigation Game



Figure 3.5 - Google Maps street view image of Bigforth Drive, the hill up to main campus (Google, 2026b)



Figure 3.6- Image of Lake Carter (Lancaster University)

In order to model the map, Blender was chosen as the 3D modelling software. Blender is an open-source, all-purpose 3D modelling software, good for a no budget project. In addition, Blender contains a free BlenderGIS plugin which can generate a 3D model from satellite imagery from OpenStreetMaps. This allowed the campus layout to be automatically generated, saving time compared to hand modelling all of the buildings. The result of this is shown in Figure 3.7.

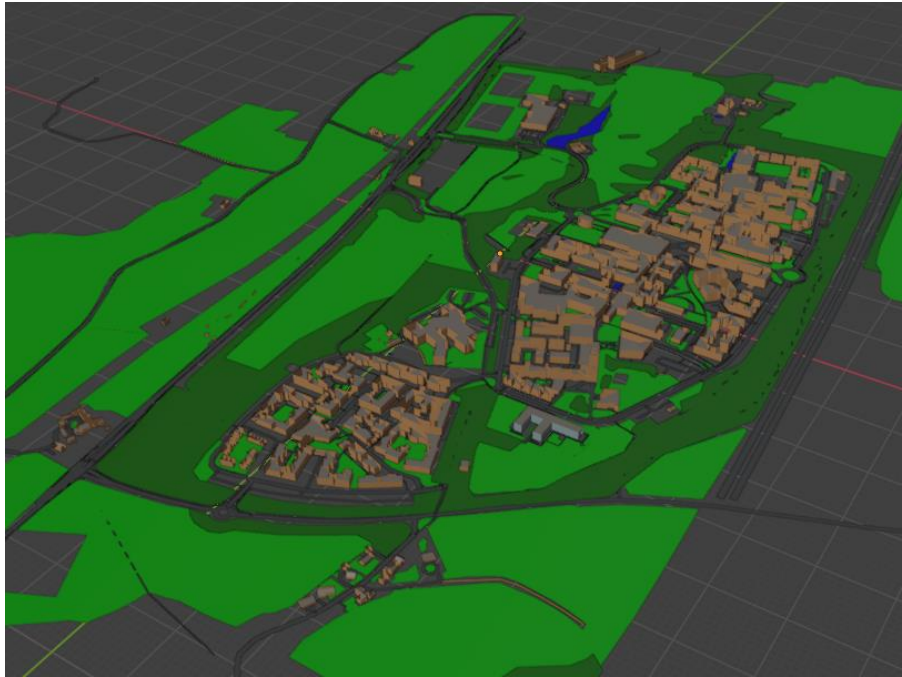


Figure 3.7 - Early campus model imported from satellite data using BlenderGIS

3.5. Game Engine

Another required design decision was what platform to use to develop the game. There were several options explored. The key decision was between a dedicated game engine, such as Unity or Godot, and a web-based approach. A web-based workflow, using technologies like HTML5, JavaScript and React, would provide great performance for the web, which would be where prospective students would play the game. With the addition of Phaser.js, a JavaScript framework for web game development, a web-based workflow would have been good for making an efficient, lightweight game. However, this approach is less successful when attempting to make a more complex game. Using a dedicated game engine allows for faster development due to the plethora of in-built tools to aid the development process. Furthermore, the performance aspect of the decision was unlikely to be an issue due to the lightweight low-poly graphics chosen for the game. Although the performance would be worse than using JavaScript with the Phaser framework, it would still meet the requirement of 60 FPS on a standard desktop PC. Therefore, due to the easier and faster development pipeline with a dedicated game engine, and the adequate performance, this workflow was chosen. The next decision was to choose the game engine. The three main options available were Unreal, Unity and Godot. However, only Godot was a viable option due to its fully open-source nature.

3.6. Restaurant Minigame

Now, the intricacies of the restaurant game loop had to be explored. As previously discussed, the game would have an open-world structure to allow for unstructured player exploration. However, a restaurant game loop was chosen as the means to keep the player playing the game. The initial idea was to take inspiration from Stardew Valley's farm game loop, where the player unlocks new crops, buildings and hence expands their farm, allowing them to earn more money. This cycle continues, providing replay-ability. For a restaurant game loop, a similar level of replay-ability was desired. Therefore, there needed to be a means for the player to improve their restaurant. This could take many forms. One idea was to allow the player to customise

their restaurant, unlocking new decorations and furniture through playing the game. However, this drew most of the focus of the game to the restaurant, and away from the key requirement of the game, which is to facilitate a spatial memory of the campus. Players spending more time inside of the fictitious restaurant would take away from the amount of exploration time. Therefore, it was decided to focus more on a delivery system. This would involve the player cooking meals, and then delivering them across the map. Therefore, more focus would be drawn to the exploration of the map, rather than spending time inside of the restaurant. The player would complete a short cooking minigame, and then would be assigned a delivery quest to deliver the meals that they cooked around the map. A minigame was chosen to provide slightly more complexity and skill, rather than only including a delivery quest. Only having a delivery quest gives limited ability for the player to improve, as all they are doing is navigating around campus. Including a cooking game also allows for more progression, not only in terms of skill, but also in terms of possible upgrades.

The specificities of the cooking minigame needed to be decided. The main requirements were that it should not distract from the core aims of the game too much, but it should provide some complexity and a skill requirement, whilst being addictive and replay-able. It should also tie into a delivery system. When exploring cooking games for inspiration, Papa's Pizzeria (Flipline Studios, 2007) stood out. Papa's Pizzeria is a popular web game, rated 4.5 stars on CoolMathGames out of 706,486 votes (CoolMathGames, 2026). It centres around a core cooking game where the player receives an order, and then has to assemble the pizza with the required ingredients (Figure 3.8), then cook the pizza for the correct amount of time (Figure 3.9), and finally cut the pizza. As you continue the game, the orders get more complicated and you have more orders to complete in less time. A similar concept could be implemented in the CNG, where the player has to assemble recipes and cook them, which is a proven concept as shown by Papa's Pizzeria. This could then be tied into the delivery system. However, the minigame would have to be simplified whilst keeping the core mechanics, as it is not the main focus of the game, unlike in Papa's Pizzeria. The idea of assembling a recipe and cooking it can be kept, but without all of the extra visual features, due to the lack of development time, and the need to keep the focus on exploration, rather than the cooking minigame. Papa's Pizzeria was chosen as the key inspiration, rather than other games like Overcooked (Ghost Town Games, 2016), mainly due to its simplicity. Overcooked is a similar cooking game where the player must assemble meals under a time limit, however it contains a great deal more complexity, with the addition of chopping, cooking, washing up, limited counter space and other features. Overcooked is also a 3D game involving movement, whereas Papa's Pizzeria is a simple 2D user interface based game. This style of game is a lot simpler and easier to implement, allowing more development time to be focused on the exploration aspect of the game.



Figure 3.8 – A screenshot of the assembly stage of Papa's Pizzeria (Flipline Studios, 2007)



Figure 3.9 – A screenshot of the cooking stage of Papa's Pizzeria (Flipline Studios, 2007)

The key aim of the cooking minigame was to be addictive and replay-able, so that the player is drawn back to the game, but also simple, so that it does not take away from the main focus, which is on player exploration of the map. Taking inspiration from Papa's Pizzeria as previously mentioned, the minigame was designed with a ticket, ingredients section and cooking area. This is a simplification of the Papa's Pizzeria game mechanics, with the assembly and cooking stages combined into one to create a simpler minigame whilst keeping the addictive aspects. This removes the mechanic of taking a customer's order to simplify the game. The assembly and cooking stages therefore needed to be designed. The assembly stage was modelled similarly to Papa's Pizzeria, with the player having to drag ingredients from the ticket onto a cooking area. Rather than having a dedicated cooking area, the player instead has

to cook each ingredient by timing a click. This is similar to the cooking area in Papa's Pizzeria, but adds more interactivity by making the player cook each ingredient. This was chosen so that the cooking minigame could include more recipes than just pizza, to give variety to the game. This would also allow more ingredients to be added, creating more exploration opportunities tying in with a quest system. Once the ingredients are cooked, the meal can be assembled. In Papa's Pizzeria, once the meal is assembled, it is handed to a customer. However, due to the requirement for the cooking game to tie into a delivery system, the meal should not immediately be given to a customer. Instead, the meals cooked should later be delivered by the player in a delivery minigame. As previously mentioned, this requirement is so that further exploration of the map is ensured.

As one of the key requirements of the cooking minigame was that it should be simple and should not distract from the core requirements of the game, it was decided that the minigame should be faster paced than Papa's Pizzeria. This would ensure that the majority of the time spent playing the game is exploring the map, rather than playing the cooking minigame. A timer could therefore be used, so that the cooking minigame never takes longer than a certain amount of time. The player would therefore have to assemble as many recipes as possible in this time limit. Due to the faster paced nature of the minigame, it needed to be responsive and engaging. After exploring fast-paced games, Fruit Ninja (Halfbrick Studios, 2010) appeared as a possible inspiration. In Fruit Ninja, the player must slice as much fruit as possible within a time limit, without slicing any bombs. This is a very successful game in the fast-paced mobile game genre, with over 500 million downloads on the Google Play Store (Google, 2026a), so could serve as useful inspiration. To make the game more responsive, Halfbrick included many visual cues in the game. These tie in with combos, which occur when the player slices multiple fruit in a row. This combo is displayed on screen, and each time the player adds to that combo, or to their score in general, a visual cue is shown, rewarding the player visually. A screenshot of this is shown in Figure 3.10. A similar concept could be applied to the cooking minigame to reward the player and make the minigame more addictive and replay-able. A combo system could be applied when the player assembles an ingredient correctly, along with visual cues to reward the player for their success. In Fruit Ninja, the combo system also influences the overall score. This score gives something for the player to improve upon, giving a reason to replay the game. A score could be implemented into the cooking minigame, based upon the combo and other metrics.



Figure 3.10 - A screenshot from Fruit Ninja showing a combo where the player slices multiple fruit in a row

Further reward could be added to the cooking portion of the game. In Papa's Pizzeria, the player is rewarded for cooking the pizza for the correct amount of time, shown in the amount of tip that they receive from the customer. A similar idea could be included in this cooking minigame. Once the ingredient is dragged into the cooking area, there could be a visual timer that shows how long it has been cooking for. The player can then be rewarded for cooking the ingredient for the correct amount of time, similar to Papa's Pizzeria. This could be adapted to suit the faster pace of this cooking minigame, by increasing the speed of the timer and adding visual feedback for timing the click correctly, as well as increasing the combo. There should also be some leniency for mistakes. In Fruit Ninja, the player can make three mistakes where they miss a fruit, but if they make more than three mistakes then the game is over. Making a mistake resets the active combo. A similar mechanic could be implemented into this cooking minigame in order to reward good play and punish mistakes fairly and with some degree of leniency. Therefore, burning an ingredient could constitute a mistake, which would reset the combo. Burning an ingredient would occur if the player does not click the ingredient in time. Therefore, another ingredient would need to be cooked. In order to make this cooking process clear to the user, colour can be used. Orange can show that the ingredient is still cooking, green can show that it is perfectly cooked, and red can show that it is burnt. This is shown in Figure 3.11. The player must therefore click the ingredient when it is green to perfectly cook it and get a bonus. So that the game is not too punishing, if the player clicks it outside of that perfect window, they will not gain any bonus, but it will still count as a cooked ingredient. However, if they leave the ingredient too long and it burns, then they must cook the ingredient again. To add a level of unpredictability to the game, ingredients could have a chance of burning straight away.



Figure 3.11 – Stages of cooking an ingredient – cooking, cooked and burnt, showing how the player must time the click to cook the ingredient perfectly

In order to add further jeopardy to the minigame and keep a high intensity, the combo could also influence the cooking time. This would involve a higher combo increasing the rate at which the ingredients cook, meaning that the player must have faster reactions. This means that the more skilful players are challenged sufficiently. To add further reward to the minigame, the player could be rewarded for completing recipes by getting time added on to the timer. This makes the timing more dynamic. The final design for the cooking minigame is shown in Figure 3.12.

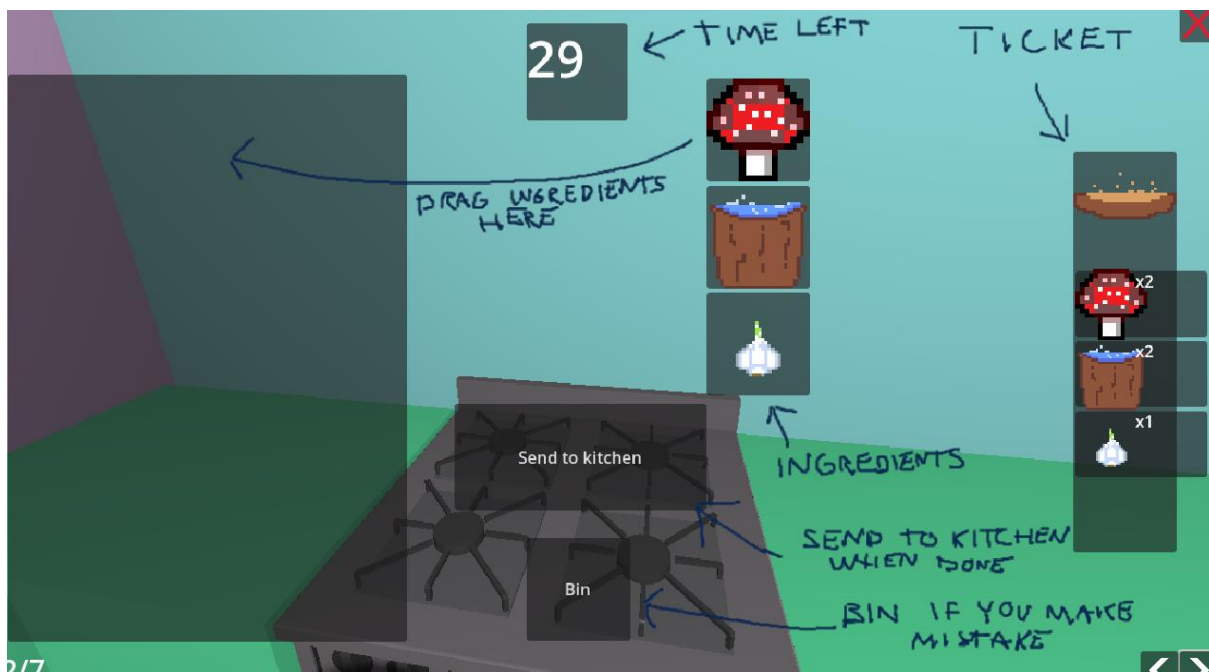


Figure 3.12 - Final design of the cooking minigame

As discussed previously, this cooking game should link to a delivery quest system. This ensures structured exploration of the map. The key requirement of the delivery system was to force players to navigate around most areas of the map, in order to build a spatial memory of the virtual campus. The delivery system was inspired by the Delivery Duel minigame in Wii Party (Nintendo, 2010), shown in Figure 3.13. This minigame involves the player delivering a pizza on a scooter across a map to a customer. They can navigate using a mini map. A similar premise could be used in the game. In order to force the player to explore more areas of the map, multiple customers could be added as destinations. In Delivery Duel, the player is provided with the exact route that they need to follow in order to reach the customer, with the minimap

acting like a GPS. As mentioned previously, Dahmani and Bohbot (2020) have suggested that habitual use of GPS negatively impacts navigational ability, as it removes the requirements to use our internal navigation system. This contradicts the core requirement of the game, which is to encourage the player to explore the map, in order to gain a spatial memory of the campus.



Figure 3.13 - A screenshot from the Delivery Duel minigame in Wii Party

There are several options available to implement a delivery quest system which meets this requirement. The first is to tell the player where the customer is located and then get them to navigate there without any assistance. The location could be described either as a place name (i.e., the customer is in Alexandra Square), or with a description of the nearest landmark (i.e., the customer is next to the tallest building on campus). Using place names was deemed ineffective, as another of the core requirements was to not focus on the map being a campus. Using campus place names would not meet this requirement. Furthermore, the focus of the game is on learning landmarks, rather than place names. Using descriptions of the nearest landmark would be an effective solution, as it meets the requirement of focusing on landmarks. However, due to the lack of landmarks modelled in detail, due to the time constraints on development, the number of locations that a customer could be would be limited. Furthermore, the player may find navigating to these landmarks difficult when they have only just started playing and have not built a sufficient spatial memory of the map yet. Another solution would be to have a pointer to where the customer is located. This could be located on the minimap or on the main screen. This would hint to the player what direction the customer is in, but would ensure that the player still has to use their internal navigation system. The minimap pointer was chosen as this forces players to translate the 2D minimap pointer into 3D navigation, aiding further spatial memory gain. Using this method also allows players to more easily navigate than using landmark names, whilst not granting as much assistance as the GPS-like system in Delivery Duel. This design is shown in Figure 3.14.



Figure 3.14 - Minimap pointer design where the red circle shows the customers location, and the house shows the player's restaurant.

Similarly to in the cooking minigame, a time limit could be added to the delivery quest, in order to force players to find the fastest route to their destination. This therefore allows room for improvement, as initially players may fail to deliver some of the meals in time. However, the timer should be longer than the timer assigned for the cooking game, as the delivery quest is the core focus of the game as it forces exploration of the map. The delivery quest should therefore terminate when the timer expires, or when the player has delivered all of the meals. Players should be rewarded for completing more deliveries. One way of doing this is to introduce a monetary system, where the player can earn money through the cooking and delivery pipeline, which could then be spent on upgrades and other things. This provides a tangible benefit to the deliveries, rather than just a score, and ties into other possible systems.

This cooking minigame and delivery quest pipeline provides a way to ensure exploration of the map whilst engaging the player. However, the main incentive to replay the minigame currently is in order to better the score, and earn more money. This is a valid incentive used in Fruit Ninja and Papa's Pizzeria, but more incentive could be introduced in a way that also encourages campus exploration. As mentioned previously, a quest system similar to that of Stardew Valley would be introduced, in order to provide some semi-structured exploration of the map. Certain features could be introduced which could tie into this quest system and provide more incentive to replay the cooking game. In Papa's Pizzeria, one way that players are persuaded to continue playing the game is through unlocking new ingredients and recipes. In that game, you unlock new ingredients and recipes through continuing to play the game. A similar progression could be added to this game. However, rather than new recipes and ingredients automatically being unlocked through playing the game, they could be unlocked through the quest system, in order to add more exploration to the game. Unlocking new recipes provides variety to the player, but could also a monetary advantage. Recipes could be assigned a cost. Therefore, successfully cooking and delivering the recipe would yield more money for the player. This would provide a more tangible benefit for unlocking new recipes. The proposed recipes are included in the Table 3.2:

Table 3.2 – List of recipes and their ingredients

Recipe	Ingredients
Beef Burger	2 bread, 1 beef, 1 tomato, 1 lettuce
Cheeseburger	2 bread, 1 beef, 1 tomato, 1 lettuce, 1 cheese
Margherita Pizza	1 pizza base, 3 tomato, 3 cheese
Mushroom Pizza	1 pizza base, 3 tomato, 3 cheese, 5 mushroom

Mushroom Soup	2 mushroom, 2 water, 1 garlic
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Researching other cooking games for inspiration on this topic, Too Many Cooks (Playstack, 2020) stood out. Too Many Cooks is a fast-paced mobile cooking game, where the player has to create dishes by chopping and cooking ingredients within a time limit, similar to in the CNG. One mechanic implemented for the sake of replay-ability in that game the use of upgrades. Players can upgrade their kitchen equipment in order to improve the cooking process. For example, you can upgrade your knife to make chopping ingredients faster. A similar system could be implemented in the cooking minigame, where upgrades can be unlocked to alter certain parts of the cooking pipeline. In order to link this to the quest system, the upgrades could take the form of different tools, which give different bonuses whilst playing the cooking game. This provides another level of progression, as unlocking new tools allows the player to reach higher scores and hence earn more money. Different tools therefore should have different characteristics that improve or detract from the cooking process. These characteristics and the reasoning are discussed in Table 3.3.

Table 3.3 – List of tool characteristics and their justifications

Tool Characteristic	Description	Reason For Inclusion
Mistakes allowed	This characteristic determines how forgiving the minigame is to mistakes, by detailing how many mistakes the player can make until they lose their combo.	Players can pick tools that are more or less forgiving of mistakes. Those that are more forgiving will provide less bonus. Including allowed mistakes makes the game feel less punishing and frustrating, but also gives the option for different playstyles – conservative or risky – adding another dimension to the game.
Burn chance	This characteristic determines the percent chance that an ingredient will immediately burn when you drag it into the cooking area.	Players can choose tools which are less unpredictable to make the gameplay less frustrating and more consistent. This is a clear motivator to upgrade tools to prevent burning, forcing further exploration to unlock these tools.
Perfect window size	This characteristic determines how large the green window is that, when clicked, gives the ingredient a perfect cook.	Players can choose tools which have a higher perfect window size, and are therefore more

		forgiving. Or, they can choose tools with a smaller perfect window size that give greater bonus. This provides a risk-reward trade-off that the player must balance.
Bonus	This characteristic determines how much combo bonus the ingredient provides when a perfect cook is achieved.	Players will aim towards unlocking a tool with a higher bonus, in order to allow them to achieve higher scores and earn more money. Tools with more bonus will have a trade-off with another characteristic.

With the format of the tools designed, different tools therefore needed to be designed. These are detailed in the Table 3.4:

Table 3.4 – List of tools

Tool Name	Description	Mistakes Allowed	Perfect Window Size	Burn Chance	Bonus
Golden Fork	Allows no mistakes in cooking but a massive 5 bonus for each correct recipe.	0	0.2	0.1	5
Knife	Massive perfect window allowed. Very forgiving.	2	0.5	0.25	2
Ladle	Beginner tool, allows lots of mistakes	5	0.2	0.25	1
Spatula	Zero chance of burning your food	1	0.2	0.0	2

The addition of unlockable recipes and tools to the cooking minigame adds more replay value due to the progression that they allow. Players now have something else to aim towards other than just bettering their score with the same variables in play. Furthermore, these unlocks tie into the proposed quest system, aiding further exploration around the map. A similar progression could be added to the delivery part of the minigame to make that more replay-able and less boring. To do this, different vehicles could be added to the game that the player can unlock. These vehicles could have different top speeds and acceleration. This would add more variation to the delivery section of the game. Initially, the player would start in a slow vehicle, but they could then unlock new, faster vehicles by playing the game, which in turn would allow them to deliver the meals faster, earning more coins. The proposed vehicles are detailed in Table 3.5:

Table 3.5 – List of vehicles

Vehicle Name	Description	Max Speed	Max Torque
Tractor	Slow vehicle that the player starts off with	400	200
Truck	Slow, but faster than the tractor	500	300
Car	Fast, top end vehicle	800	400

Given the use of vehicles in the delivery quest, the routes that players take need to be considered. This involves considering where the player starts the delivery, and where the customers are located that they need to deliver to. One idea would be to spawn the player in a random location on campus each time they enter the delivery portion of the game. However, due to the use of vehicles, the player should not be encouraged to drive in pedestrian areas of the map. Instead, the player could be spawned at the Underpass each time. The Underpass is a section of road that runs underneath the central Alexandra Square on the Lancaster University campus, and is where all buses run through. In order to further encourage the player to stick to the roads, rather than drive through pedestrian areas, the possible customers could be situated only around the perimeter roads on the map. Therefore, the best route would always be to drive on the roads. This is important as these habits should then transfer over into the real-world campus. If the player is used to driving on pedestrian areas of the virtual map, then they may get confused when first encountering the real-world campus.

3.6. Other Mechanics

With the core game loop of a cooking minigame and delivery system designed, more elaboration was needed on the other parts of the game. As detailed previously, much of the inspiration for the game comes from the open-world quest system of Stardew Valley, due to its ability to encourage exploration of the map. This quest system needed designing. In Stardew Valley, quests are triggered by interactions with NPCs. This gives the game a sense of life, as these NPCs populate the game map, making it feel full. NPCs could be included in the game for the same reason. Certain NPCs could give the player quests, adding to the structured exploration of the map. These quests could also link to the unlock system. Certain quests could reward the player with new tools, recipes or vehicles. This would give a reason for the player to complete the quests. Other quests could give the player money as a reward. In Stardew Valley, the player can view their quests in a log, which allows them to keep track of their

progress. With the inclusion of quests into the campus navigation game, the player needs a similar way to be able to track quest progress. There should also be rewarding visual feedback when a player completes a quest, such as a pop up congratulating them and detailing what they receive from completing the quest. This provides a positive feedback loop, which encourages the player to continue completing quests.

In *Stardew Valley*, another way to unlock items is through shopping and trading. There are shops around the map which the player can purchase items from. This mechanic could also be included in the game, in order to add more variety to how players unlock items. Shops could also be adapted to encourage further exploration, as they could be placed around the map so that the player has to explore the map in order to find them. With the addition of trading, players now have two ways of obtaining items, through trading and through quests. Both of these mechanisms provide encouragement for players to explore the map, whether directly through quests or indirectly through trading. Another possible mechanism to encourage exploration is that of crafting. *Minecraft* (Mojang, 2011) includes a crafting system whereby the user can use a crafting table to craft new items from other items. The crafting table takes the form of a 3x3 grid which the user can drag items into, shown in Figure 3.15. In *Minecraft*, this encourages exploration as players must search for the desired items in order to craft the recipe that they require. A similar mechanic could be implemented in the campus navigation game. This could take the form of a 3x3 crafting bench, similar to *Minecraft*. This crafting bench could exist inside of the restaurant. This would further establish the restaurant as a base of operations for the player at the centre of campus, and mean that they must navigate back to the centre of campus each time they want to craft something. This further encourages navigation of the map, building the spatial map more. The crafting system could then tie into the trading and quest system, as all three systems would rely on items to function.



Figure 3.15 - A screenshot showing the crafting grid and inventory from *Minecraft* (Mojang, 2011)

Another key feature of *Minecraft*'s inventory system is the ability to see the crafting recipes – how to create each item. Without this, players would have to guess how to craft each item.

Therefore, it was key to add this to the crafting system. The final design of the crafting bench is shown in Figure 3.16.

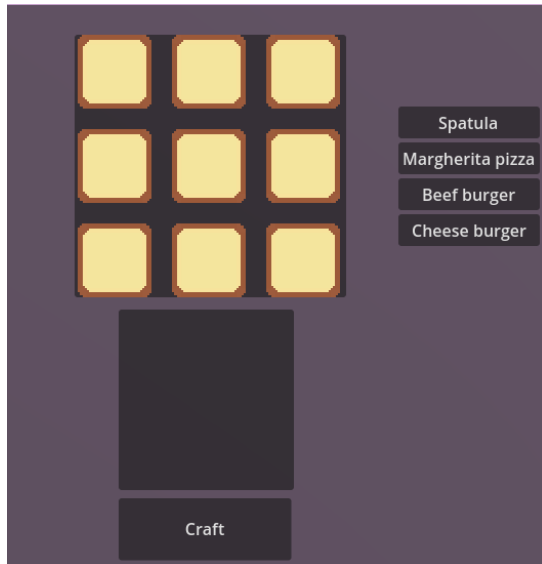


Figure 3.16 - The final design of the crafting bench

In order to implement a crafting and trading system, whereby the player can purchase, trade for and craft items, an inventory system needed to be designed so that the player can carry and keep track of their items. Minecraft (Mojang, 2011) implements this system, as shown in Figure 3.15, with a 3x9 inventory and 1x9 hotbar. The hotbar is always showing in the UI, whereas the inventory is toggled with the press of a button, as shown in Figure 3.17. Stardew Valley (ConcernedApe, 2016) implements a similar inventory system, with a hotbar and inventory separated. In both of these games, the hotbar is separated as you have the ability to use your items in the world. However, in the campus navigation game, items are only used to craft unlocks which are used in the cooking game and delivery game. Therefore, a separate hotbar is unnecessary and only leads to visual clutter. However, the inventory itself is a necessary asset in order to facilitate the storage of items. The key feature of both the Stardew Valley and Minecraft inventory systems is the ability to stack items, as shown in Figure 20. This means that the inventory does not get immediately filled up with the same item. For the inventory in the campus navigation game, less slots are needed, due to the relative lack of items compared to Minecraft and Stardew Valley. Therefore, only a 2x5 inventory was designed, shown in Figure 3.18.



Figure 3.17 - A screenshot of the hotbar in Minecraft (Mojang, 2011)

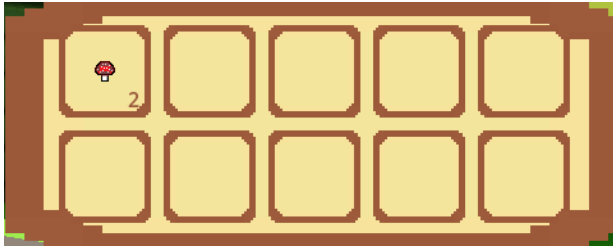


Figure 3.18 - The final design for the inventory, showing an item being stacked.

3.7. Game Rules

Overall, based on research on other games and considerations of what game design decision best fit the requirements, the final game design rules are as follows:

Map

- Must be open-world.
- Players must be able to freely roam the map with no constraints.
- Must have a 3D first person perspective.
- The map must be a representation of the Lancaster University campus.
- The map must contain the key landmarks as discussed.
- The map must be populated by NPCs

Cooking Minigame

- There must be a drag and drop cooking minigame.
- The players must drag ingredients into a cooking area and then time clicks to cook the ingredients correctly and complete the recipe.
- Players must assemble the correct ingredients and cook them to finish a meal.
- There must be a timer.
- There must be satisfying visual feedback.
- There must be a combo system.
- Must have unlockable tools which influence the cooking minigame
- Tools should have 4 characteristics: Mistakes Allowed, Perfect Window Size, Burn Chance and Bonus.
- Must have unlockable recipes which influence the cooking minigame
- Recipes should have different ingredients and should reward the player with more or less money when cooked.

Delivery System

- The cooking minigame must segue into a delivery game.
- Players must be spawned into a vehicle at the Underpass.
- They must then be assigned customers to deliver to.
- These customers must be represented by red dots on a minimap.
- There must be a timer for the player to deliver the items.
- The customers must be located on roads.
- Must have unlockable vehicles which influence the delivery game

Quest System

- Some NPCs must give the player quests to complete
- These quests must encourage exploration of the campus
- Must have a quest log which is a visual tracker of quest progress.

Crafting System

- Must have an item driven crafting system where players can craft some tools, recipes or vehicles from items

Trading System

- Must have an item driven trading system where players can purchase items, tools, recipes and vehicles from certain NPCs.

Inventory

- Must have an inventory system which allows players to hold items on their person.

3.8. Subsystem Design

The following section will detail the key subsystems required to implement the game rules as set out above.

Quest Subsystem

A modular quest subsystem is required. The design is shown in Figure 3.19. The QuestManager is a single entity that creates, completes and manages quests. Therefore, NPCs and players can interact with this to easily create and complete quests. It also allows a modular approach to UI, as the UI can use an event-driven approach to be notified when there are any updates. Quests were added as a custom resource, with an ID to allow ease of management. Therefore, NPCs could interact with the quest manager to assign quests, and UI elements could interact with the quest manager in order to display quest status. Quests themselves were developed as a custom resource, containing relevant information and a list of objectives. Objectives were also implemented as a custom resource. Therefore, one quest can have multiple objectives, allowing for multifaceted, more interesting quests. The UI involves a short quest journal on the main UI, and a more detailed quest UI if the journal was expanded. This gives the player the ability to see all of their active quests and their progress towards said quests, meaning that they can more

easily keep track of what they need to work towards. This should also be created modularly, with quest entry scenes developed to hold quest information such as progress, description and title. Alongside this, a UI pop up was added to display when a quest was completed, giving information about the quest and the reward gained. This was done to give a further sense of reward when quests are completed.

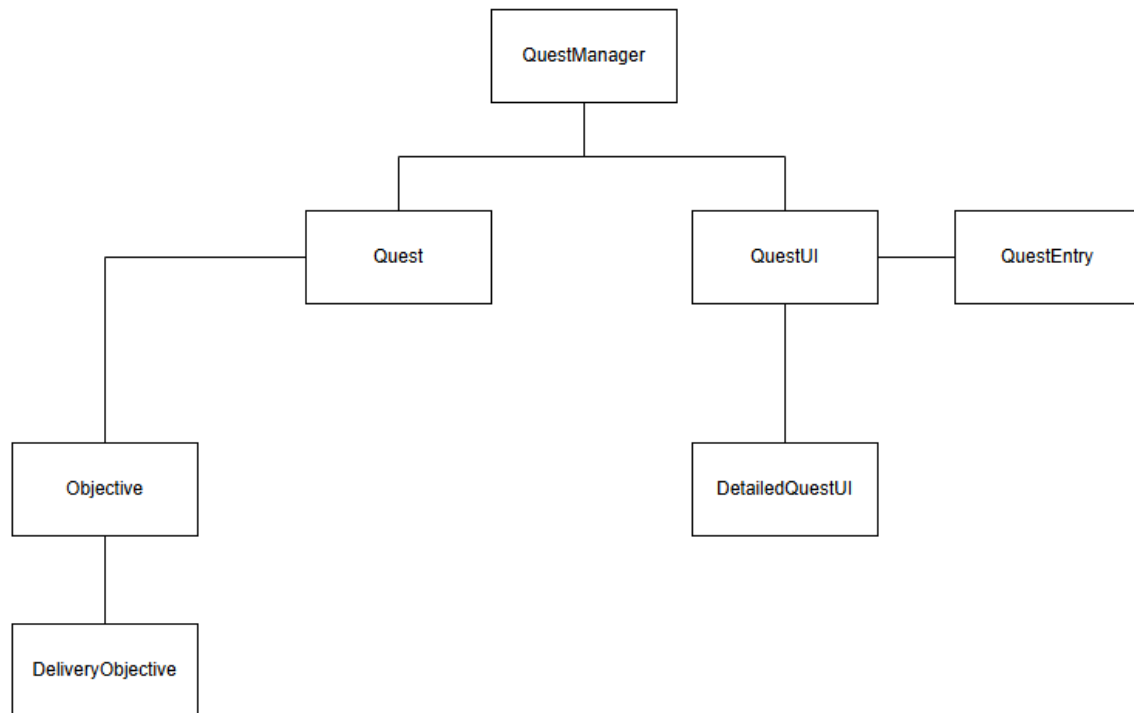


Figure 3.19 - Quest Subsystem Diagram

NPC Subsystem

The NPCs can be split into two categories: patrolling and trading, as shown in Figure 3.20. Patrolling NPCs can give quests and patrol the map. Trading NPCs handle trading and purchasing items. These trigger the shop UI. This would consist of shop entries, containing an item and its price or trade.

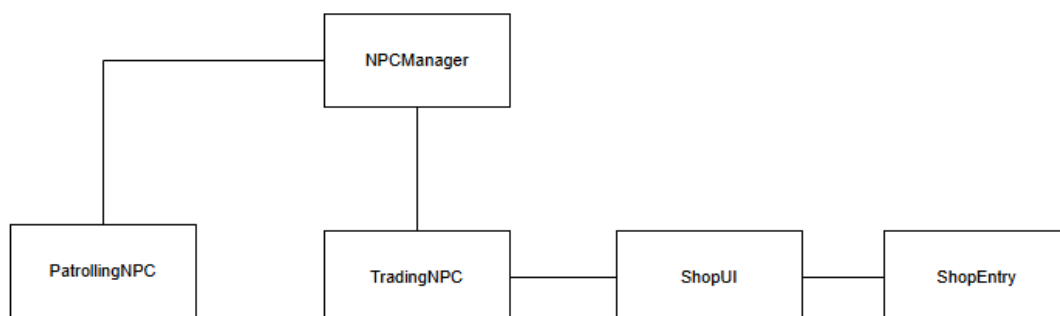


Figure 3.20 - NPC Subsystem Diagram

Crafting and Inventory Subsystem

The design for this is shown in Figure 3.21. Entities can contact the InventoryManager to put or take things from the inventory. The CraftingManager links to this, handling crafting. This provides a modular system and separates functionality from data. Inventory and CraftingRecipe can hold the data. InventoryItem is a modular custom resource which contains an item. This can be used across the game.

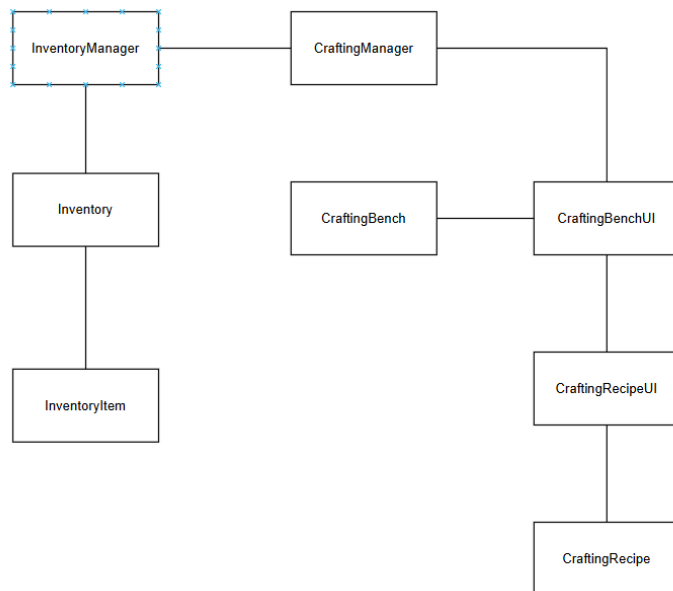


Figure 3.21 - Crafting System Diagram

Player Subsystem

The player subsystem can be split into two main parts, as shown in Figure 3.22. The player controller can handle the movement, collisions and camera. The player data manager can handle data held about the player, such as coins, tools owned, etc. The core concept being that the game should reward players for completing quests in order to incentivise them to continue with the game and traverse the campus map even more. Therefore, a player stats system was implemented, with coins and reputation being the two key metrics that a player can gain. Coins were introduced in order to give an economy to the game and allow for players to purchase items later down the line. Reputation was added as a measure of how popular the player is among the townsfolk, to incentivise improvement and as a reason to spend money. These metrics were added to the UI in order to allow players to keep track of their stats whilst playing the game.

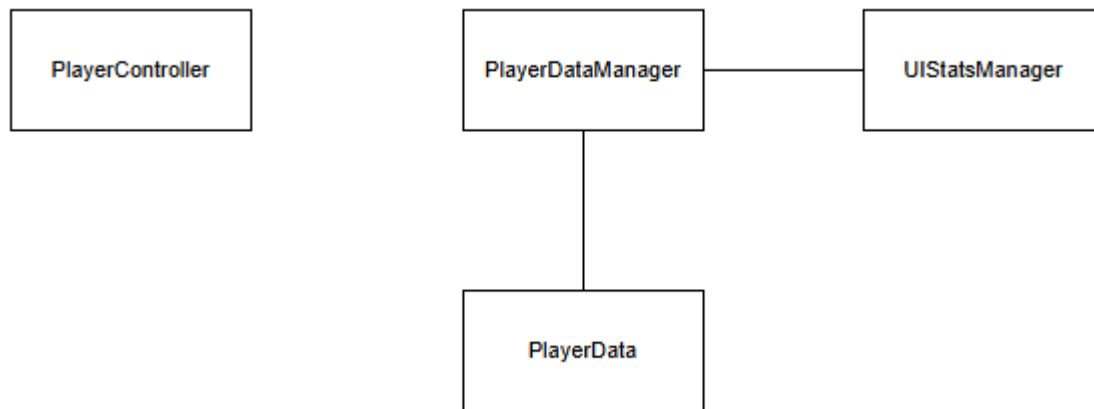


Figure 3.22 - Player Subsystem

Cooking Minigame

The cooking minigame can be modelled as shown in Figure 3.23. This models the drag and drop design as set out earlier in the chapter, whilst keeping it modular with a recipe custom resource depending on an item.

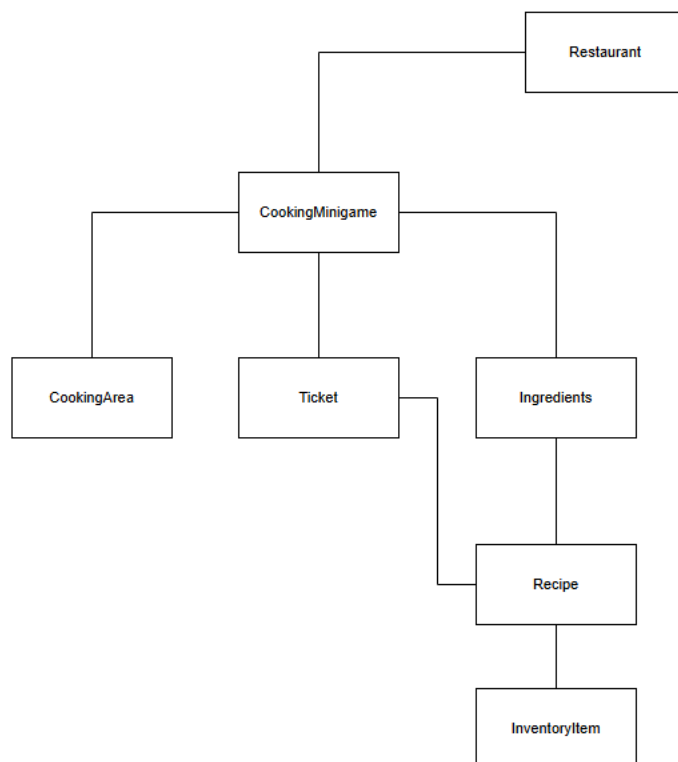


Figure 3.23 - Cooking Minigame Subsystem

Vehicles

The aim for introducing vehicles was to create a modular implementation, so that multiple different vehicle types could be added to the game under the same template. In order to achieve

this aim, two things were must be implemented – a vehicle scene and a vehicle custom resource. The vehicle scene being the basic template of the 3D vehicle that each different vehicle could inherit from, and the custom resource containing the vehicle statistics such as the acceleration, max speed and weight. Crucially, this acts to separate the data and the active implementation of the vehicles.

3.9. Data Design

In order to make the videogame modular, much of the data can be modelled as custom resources.

- Quest
- Objective
- Delivery Objective
- Recipe,
- Vehicle
- Inventory item
- Shop entry
- Crafting recipe
- Tool

4. Implementation

4.1 Process

Git version control was used in order to maintain the structure of the project and keep track of progress. The project was clearly organised into folders, with Godot scenes all kept in one folder, all scripts kept in another folder, and separate folders created and maintained for other game assets. Where possible, modularity was maintained in the project, reusing scenes, scripts and classes.

The implementation process followed the game rules and subsystems set out in the design. Initially, a basic, untextured campus map was created from satellite data. This prototype was introduced to allow other features of the game to build upon it, without having to wait for the development of a fully-fledged campus map, the design of which would take many weeks. With a basic campus map in place, the player controller subsystem was implemented, to allow the player to move around the map. The first subsystem to be developed was the inventory system. The reasoning for this decision was that this made up the foundation of many of the other subsystems outlined in the design of the videogame. Many further systems depended on the use of items, which were defined as a custom resource to allow for reuse across the game. For example, crafting, trading, cooking and quest systems all require the use of items in some way. Along with defining items, the inventory subsystem involved the development of an inventory and the ability for players to pick up and stack items in their inventory, which would be needed later on for when players need to collect items, buy items or receive items from quests.

Once the inventory system was completed, the NPC subsystem was developed. Initially, the patrolling NPCs were coded to follow a set path. However, this functionality was too stiff and unrealistic. Therefore, the NPC code was updated to provide a more dynamic, interesting behaviour. The NPC was given functionality to patrol between points, chase players, and idle. Therefore, NPCs could be set to patrol around certain areas of the map, walking between the points, and then idling. Furthermore, certain NPCs could be set to follow the player, useful to force the player into interacting with certain NPCs. Overall, this behaviour made the NPCs feel more alive and responsive, adding to the immersive experience of the game. With the NPC movement completed, an interaction system was needed, to allow players to interact with the NPCs and start and complete quests. Therefore, a modular interaction system was developed. Rather than develop an interaction system solely for use with the NPCs, a broader system was developed, with an interactable and an interacting component. Therefore, with the player as the interacting component, anything could be assigned as an interactable, whether that be an NPC, an item (to pick up), a trader, a door, etc. The interactable was designed so that a text prompt would display when the player is within a certain range, displaying instructions on how to interact. For the NPC, they should display text when interacted with, either to represent friendly conversation (to make the campus feel friendly and alive), or to work with the quest system.

Therefore, a dialogue manager was created next. This allowed dialogue to be displayed and generated in an interactive way on the screen, with a text scroll effect and the ability to display multiple lines of dialogue separately. This was implemented in order to make the NPCs feel more alive, rather than simply displaying necessary information as a large block of text. Once the dialogue system was implemented, NPCs now had a way to display text. This meant that they could display remarks as if they were talking to the player. When developing the dialogue system, a bug was encountered where the system would freeze if the dialogue line was too

short. This was due to the code awaiting a resize of the text bubble, which did not happen if the text was too short. A proper fix was attempted, but did not work. Therefore, in order to prevent spending too long on this bug, all dialogue lines had to be longer than this short limit. The next step was to implement the quest subsystem, which is one of the founding ideas in this game. This would allow NPCs to give, manage and complete quests. With the quest system implemented, a rewards system was required, in order to make the quests worth completing.

With the monetary system implemented, a trading system could be introduced, allowing players to spend the coins that they have earned. Therefore, the trading NPC was created. These NPCs are stationary, but can be interacted with, which brings up a shop in which players can purchase or trade for items on sale. This was implemented as a shop UI with shop entries as custom resources, each containing an item, quantity, price and other key information. Players could then purchase items using their coins.

With these key subsystems implemented, the core game loop could be concentrated on. As mentioned in the design section, this key game loop was the restaurant and delivery system. Firstly, a separate restaurant scene was implemented, with a door to allow players to enter the restaurant from the main scene. Then, the cooking minigame was implemented. A recipe custom resource was created, with ingredients, output item and quantities. With this, a drag and drop game was created, whereby a recipe ticket is displayed to the player and they have to drag the correct ingredients into a cooking area and cook the ingredients by timing clicks in certain ways. Once all ingredients have been added and cooked, the recipe can be submitted and a new recipe will appear. To make the game more satisfying, combos were added, along with visual feedback, to reward good play. Once all recipes are cooked, the cooking minigame finishes. The next step of development was to link this cooking system to a delivery system whereby the player must deliver the recipes that they have cooked across the campus map. Initially, this was facilitated by creating a new delivery objective class that inherited from the original objective resource. This class allows delivery objectives to be created, which contained the ID's of NPCs to deliver to. In order to visualise the delivery stage of the game, a minimap was introduced. This contained the restaurant location, location of the player, and the locations of the NPCs which the player needs to deliver to. Several issues arose with this system, including the issue of the main scene refreshing when the player exits the restaurant scene. To tackle this issue, the NPC manager script was used to link NPC IDs to their respective nodes in the active scene. Another issue was that players were not prevented from playing another cooking minigame before they had finished the delivery. This meant that multiple deliveries could be opened at once. This was fixed by preventing the player from starting the cooking minigame if a delivery was active.

The next step in development was to introduce the customisation and crafting systems. First, the customisation menu was added. This aimed to contain the unlockable items that a player could obtain – namely new tools, recipes and vehicles. These three unlockables were added as custom resources to allow for multiple of each type to be later added to the game. The player data singleton was also added to keep track of the different unlocks that the player had, as well as which unlocks were selected. This was displayed on a UI, accessible inside of the restaurant. The template of a crafting bench was also added to the restaurant scene, with the idea of having all of the customisation options within the restaurant scene, intending for the restaurant scene to act as a home base for the player.

Next, the delivery system needed extending to add vehicles as per the design. With the vehicle templates created, this was then linked to the delivery system, whereby the player was spawned into a vehicle once they exit the restaurant, and the vehicle despawns once the timer expires, or the player has delivered all of the food.

The next subsystem on the itinerary was the crafting subsystem. This aimed to make use of some of the code for the inventory system implemented prior, such as the inventory slots, inventory items, and others.

With the crafting system implemented and the player now able to craft different tools and recipes, all of the core subsystems were created. The next step in implementation was to add any extra features, fix bugs and polish the game. The initial step was to test the existing functionality for the core subsystems already implemented. Doing so led to many bugs being located. This was due to the subsystems being implemented relatively quickly and without a final layer of polish. This development methodology was chosen so that all of the core functionality of the game would be present before the deadline. If all of the subsystems were perfectly tweaked before moving onto the next system, then the project risked running out of time. Instead, all of the key systems were implemented in a basic state. The next stage of the implementation was to fix these bugs that were located in the core subsystems. Once these severe bugs were fixed, time was assigned to adding a layer of polish to the game, and adding more quality of life features to make the game more enjoyable.

One of the key polish features added in the latter stages was animations. These were added to the NPCs and the player for walking, jumping, running and sitting in order to make the movement within the game more realistic and interesting. These animations were retrieved from mesh2motion under an open source license. Initially, original models and animations were going to be created, however due to the time pressure, completing all of the core functionality of the game was prioritised over custom models and animations.

With a level of polish added, the focus then shifted to populating the game. This involved positioning NPCs around the campus environment. It also involved creating the different resources for the game. Firstly, this was creating all of the unlockable tools, recipes and vehicles that the player can work towards unlocking during the game.

The next step was to model the campus map in more detail, taking it from the template from earlier in development to the desired quality as set out in the design. This was done in Blender.

4.2. Testing

One of the key requirements of the CNG is that it runs with at least 60 frames per second (FPS) on a desktop PC with 16GB RAM, 6-core processor, and mid-range dedicated GPU at 1080p. To test this requirement, MSI Afterburner and RTSS were used. The test system specifications were as follows:

- CPU: AMD Ryzen 5 3600
- RAM: 16GB DDR4 3200MHz
- GPU: AMD Radeon RX 5700 XT
- Display Resolution: 1920x1080
- Operating System: Windows 11

These specifications match the ones in the requirements section, with all components being at least 5 years old. This represents a realistic desktop system which a student would own. Testing was performed across multiple scenarios to represent typical usage of the CNG.

Scenario 1: Idle. The player is idle, not moving or moving their camera.

Scenario 2: Cooking and delivery minigame. This tests the performance of the core game loop as a whole.

Scenario 3: Exploring. This tests the performance when the player is exploring the open-world.

Scenario 4: Cooking minigame. This tests the performance of the cooking minigame on its own.

Scenario 5: Delivery game. This tests the performance of the delivery game when the player is delivering meals across the map.

For each scenario, the CNG was run for 3 minutes. Average FPS, minimum FPS, maximum FPS, 1% low and 0.1% low FPS. The results are shown in Table 4.1.

Table 4.1 – FPS statistics for game scenarios

Scenario	Average FPS	Minimum FPS	Maximum FPS	1% Low	0.1% Low
Idle	140.6	119.5	145.4	50.7	45.5
Cooking and delivery minigame	117.6	0.6	145.9	30.9	0.6
Exploring	122.8	79.3	145.8	49.1	39.1
Cooking minigame	143.9	142.9	144.8	137.2	132.2
Delivery game	106.7	25.4	145.6	44.1	31.5

The system maintained an average frame rate above the target 60FPS in all tested scenarios. However, in the high load delivery game, the minimum FPS dropped below the target. Furthermore, the 1% and 0.1% lows also dropped below the 60FPS target. This suggests that the requirement is largely satisfied during typical gameplay. However, during exploration and the delivery game, performance degrades. This is likely due to the number of models in the main map, leading to greater rendering and processing demands. The FPS dropped significantly when exploring dense areas towards the centre of the map. Optimisation techniques such as Level of Detail (LOD) could be implemented, which would swap distant models for lower-poly versions. This would reduce GPU load. The 1% and 0.1% lows dropped below the target. These metrics measure the average framerate of the slowest 1% and 0.1% of frames respectively. This provides a measure of stutter and consistency. The low values of these metrics indicates stutter present in the gameplay. The value of 0.6 for the cooking and delivery minigame represents switching scenes from the restaurant to the main scene, however the other values occurred during normal gameplay. This could be due to increases in rendering load, collisions or inefficient scripts.

Overall, this testing suggests the 60FPS requirement was largely met, as during all of the tested scenarios, the average FPS was significantly more than 60FPS. However, in high-load scenarios, minimum FPS and 1%/0.1% lows dropped below the target, meaning that the game is not sufficiently smooth or optimised. Due to this, the requirement was not fully satisfied.

5. Evaluation

5.1. Design

The evaluation was designed to test both the usability of the CNG, and the effect that it has on campus navigational ability and landmark recognition of the campus, as these were the core requirements of the system. If put into production, the CNG would be given to prospective university students who are relatively unfamiliar with the campus. Therefore, study participants who are relatively unfamiliar with the campus needed to be recruited, to model the navigational ability of an incoming student. The requirement that the CNG map should not focus on being a campus (the implicit learning position) was considered a secondary requirement and therefore out of scope of this study.

The evaluation needed to be designed in order to evaluate the core requirements of the CNG. These were that it should foster landmark recognition, and should improve navigational ability of the campus. In order to meet this criteria, photos of the 11 key landmarks around campus modelled in the CNG were taken in real life. These were the same landmarks that were modelled in detail in the game. Then, 8 routes across campus were plotted on a screenshot of the OpenStreetMap of Lancaster University. These routes were plotted so that they would pass several of the key landmarks. The core idea is that the participant would have to identify which landmarks you would pass on each route. This would test the landmark recognition requirement of the CNG. The routes would be split, so that 4 routes would be tested before the participant played the videogame, and 4 would be tested after they played the videogame. Differences could then be observed to evaluate whether the videogame has any effect on the navigational ability of the participants. The routes were general routes spanning most areas of the campus, designed to test the general spatial memory of participants before and after playing the CNG. An example route is shown in Figure 5.1.

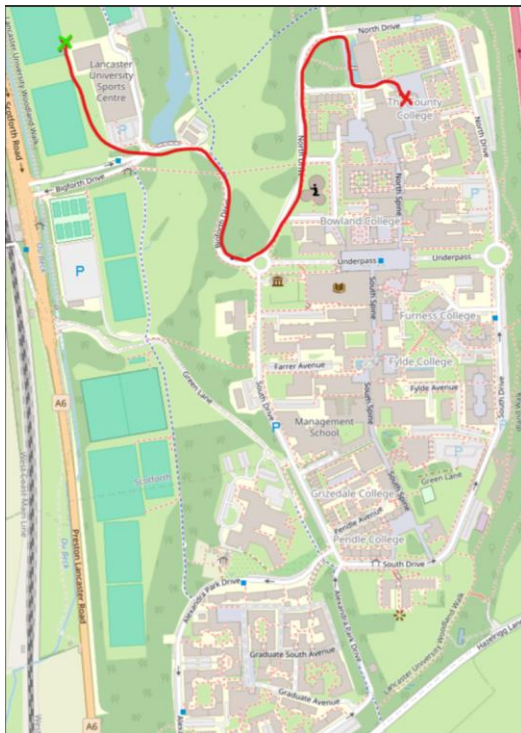


Figure 5.1 - Route 1 from the evaluation

In the pre-game activity, the participant therefore had to identify which of the 11 landmarks you would in routes 1 to 4. They were given 2 minutes for each route, but could finish early if they believed they had identified every landmark. This time was recorded in order to compare to the post-game activity, to assess the speed of landmark recognition. As well as the time taken, the precision and recall should be recorded, in order to monitor the accuracy of the responses. Precision measures the accuracy of positive predictions – in this case, of the landmarks that the participant selected, which were actually present on the route. Recall measures the ability to identify all positive cases – in this case, of all the landmarks that were present on the route, how many did the participant correctly identify. In order to simulate the experience of a prospective student, they could also use Google Maps and MazeMap, the use of which was also recorded. As participants were expected to have little to no knowledge of the campus, they were initially expected to struggle with the navigation task, similar to the experience of an incoming student. Therefore, they were granted access to the GPS tools of Google Maps and MazeMap, the same as an incoming student would have access to. As well as recording accuracy metrics and time taken, user feedback was also taken into account. Participants would be asked how they felt the pre-game activity went.

Although the metrics of precision, recall, time taken and user experience may provide evidence of a change in navigational ability from pre-game to post-game, more information would be needed in order to understand how this change in navigational ability would come about. Therefore, participants were asked to voice record themselves narrating their thought process when completing the activities. This information should show how the participants are identifying the landmarks from each route. If the CNG has a positive effect, then you would expect for participants to initially rely heavily on GPS-based tools such as Google Maps and MazeMap, but after playing the game would recognise landmarks without having to use these tools.

The pre-game task was designed to test how the participant handled a navigation task as a baseline. The participants were expected to have little to no initial knowledge of the campus layout and therefore where each landmark would be situated. Therefore, they should be fully reliant on the use of GPS tools to help them identify which landmarks occur on each route. Participants would be expected to see the route, and then look at the GPS tool and match the buildings to the landmarks in the photographs. This could be done by matching the expected shape from the 2D route map to the photograph, or via Google Street View, which shows a 3D view of the campus.

This evaluation was designed in order to get participants to take part in a task which closely matches the type of navigation they would have to do if arriving on the real-world campus for the first time. Ideally, the effects of the videogame on real-world campus navigation would have been directly tested through in person navigation tasks. However, getting participants who had never been to the Lancaster University campus to travel to the campus and complete tasks in person was infeasible and out of the scope of this project. Therefore this evaluation task was designed to test participants' navigation skills in a similar manner to a real-world task. This involves testing their landmark recognition and their ability to recognise landmarks along a route, which is key when navigating a campus to reach lectures and other destinations in real life.

Once this activity had been completed, participants were asked to play the CNG for 10 - 15 minutes. This time was chosen as it should allow participants to substantially explore the map, whilst not taking up too much participant time. Participants were given the videogame and no instruction, so could play however they desired. The game points the player to start a delivery, but as it is open-world, this is not enforced. If the game achieves its aims, then playing the

videogame should cause the participants to improve their navigational ability for the campus, and improve their landmark recognition skills for the campus, by developing a spatial memory. These skills should translate from virtual navigation within the game to real-world navigation on campus physically, and also semi-real-world navigation in terms of the evaluation task of identifying landmarks from routes plotted on a map.

After playing the game, the participants were asked to complete a post-game activity. This was in the same format as the pre-game activity, but looking at routes 5 to 8 now. Precision and recall were recorded again, as well as the time taken. The participants were then asked how confident they were in that activity and whether they used external tools. The participant was also asked to narrate their thought process for this task in a voice recording. This task was designed to test how the participant's thought process and navigational ability changed after playing the videogame, especially considering landmark recognition as a key facet of navigation. If successful in fostering landmark recognition and navigational ability improvements, then the participants should be able to identify the landmarks along each route with more accuracy, consistency and speed, and with less reliance on the external tools of MazeMap and Google Maps. This would suggest that the videogame is effective in fostering landmark recognition and navigational ability of the Lancaster University campus, and therefore would be an effective tool in reducing the cognitive load on new students entering the campus for the first time.

The participants then had to fill in a short questionnaire about their experience, including a System Usability Score (SUS) questionnaire. These questions were designed to evaluate the usability and enjoyment factor of the videogame. This is an important evaluation as a key part of learning is repetition. If students are going to retain navigational knowledge and landmark recognition for the campus long term, then repeated exposure to the navigational tasks within the game is necessary. Although playing the game once or twice may improve navigational ability in the short term, repetition will help to foster a long term understanding. Therefore, replay-ability is a very important trait for the game to have. This is one of the traits evaluated from this questionnaire. If participants feel that they would continue to play this game for an extended period of time, then this suggests a great level of replay-ability. Another of the key requirements of the CNG is that it should not cause confusion. This is also a factor assessed by a SUS score.

5.2. Results

In order to measure the accuracy of the landmark recognition, three metrics were calculated – the mean F1 score, mean precision and mean recall. These were calculated for the pre-game task, and for the post-game task, to see the impact of playing the videogame on landmark recognition accuracy. The results are shown in Figure 5.2. The first metric was mean precision. The mean precision for the pre-game activity was 0.938. The mean precision for the post-game activity was 0.958. This is an increase of 0.02. Precision measures the accuracy of positive predictions – in this case, of the landmarks that the participant selected, which were actually present on the route.

The next metric recorded was mean recall. The mean recall for the pre-game activity was 0.638. The mean precision for the post-game activity was 0.683. This is an increase of 0.045. Recall measures the ability to identify all positive cases – in this case, of all the landmarks that were present on the route, how many did the participant correctly identify.

Campus Navigation Game

The next metric recorded was F1 Score. This is the harmonic mean of precision and recall. The mean recall for the pre-game activity was 0.730. The mean precision for the post-game activity was 0.783. This is an increase of 0.053.

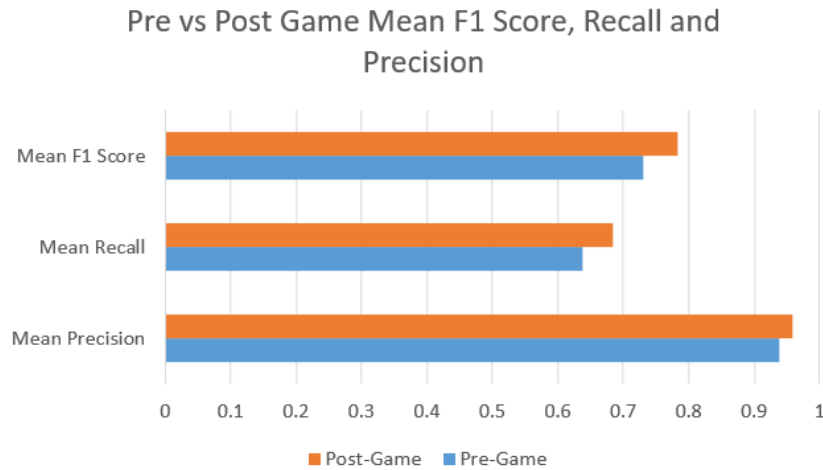


Figure 5.2 - Mean F1 Score, Recall and Precision comparison from Pre-game to Post-game

The mean time taken for the pre-game and post-game tasks was also recorded and is shown in Figure 5.3. The mean pre-game task completion time was 127 seconds. The mean post-game time was 124 seconds. This is a decrease of 3 seconds.

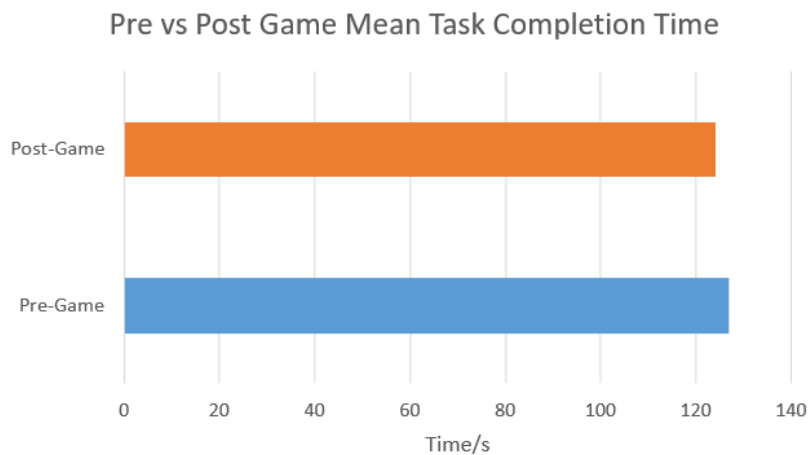


Figure 5.3 - Mean task completion time comparison from Pre-game to Post-game

Participants were also asked to rate their confidence on the post-game and pre-game tasks. This is shown in Figure 5.4. The mean pre-game rating was 2/5. The mean post-game rating was 3/5.

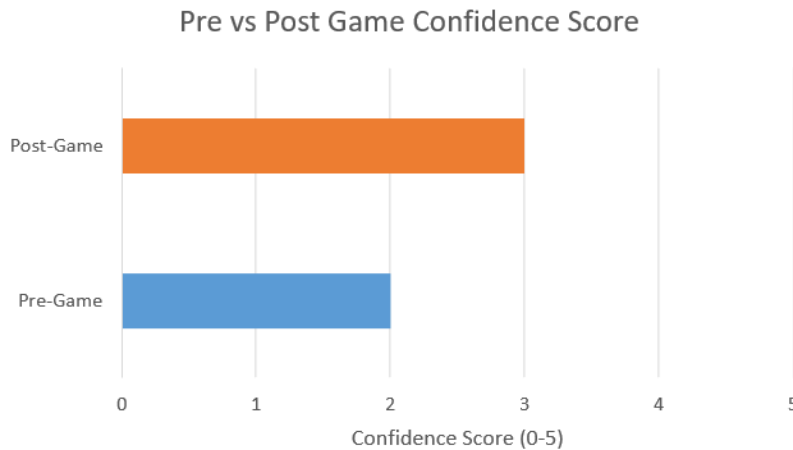


Figure 5.4 - Confidence score comparison from Pre-game to Post-game

The number of false negatives for each landmark was also recorded, and is shown in Figure 5.5. This refers to how often each landmark was classified by a participant as not in a route, when it actually was. Landmark 1 (LICA), Landmark 4 (Bowland Tower) and Landmark 10 (Underpass) were the most frequent false negatives. Conversely, Landmark 3 (Alexandra Square), Landmark 6 (Fylde Accommodation), Landmark 7 (InfoLab), Landmark 8 (Sports Courts) and Landmark 11 (Pendle Pillars) had no false negatives.

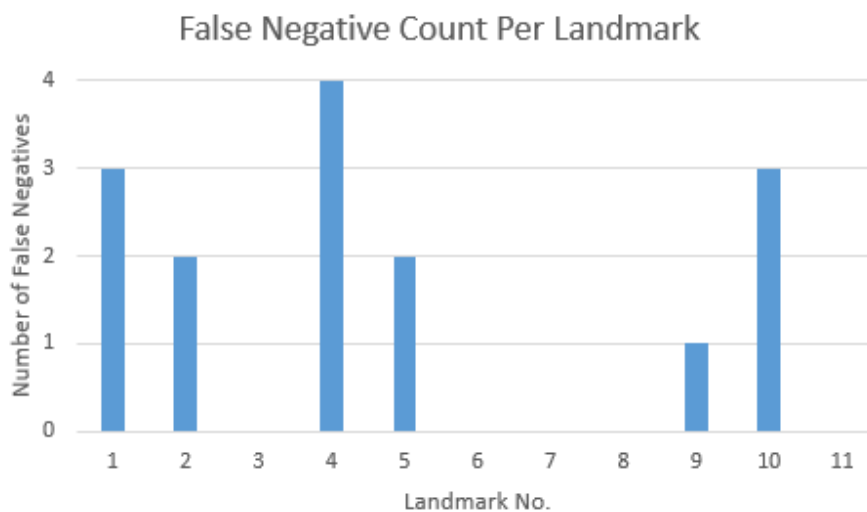


Figure 5.6 - False Negative Count Per Landmark

Participants were also asked whether they relied less or more on external GPS tools, such as Google Maps or MazeMap. All participants answered that they relied less on them. Participants were also asked whether they thought that playing the videogame improved their ability to navigate the campus, and all agreed that it did. When asked why, the common theme was that it helped them become more familiar with the shape of the campus, and certain landmarks.

Participants were also asked to undergo “think aloud” narration whilst they completed the pre-game and post-game tasks. This revealed that participants relied on Google Maps. In the post-game tasks, they sometimes recognised landmarks without having to consult Google Maps.

A System Usability Score Questionnaire was also provided to the participants after completing all of the tasks. The mean SUS score calculated is 50. This is shown in Figure 5.7.

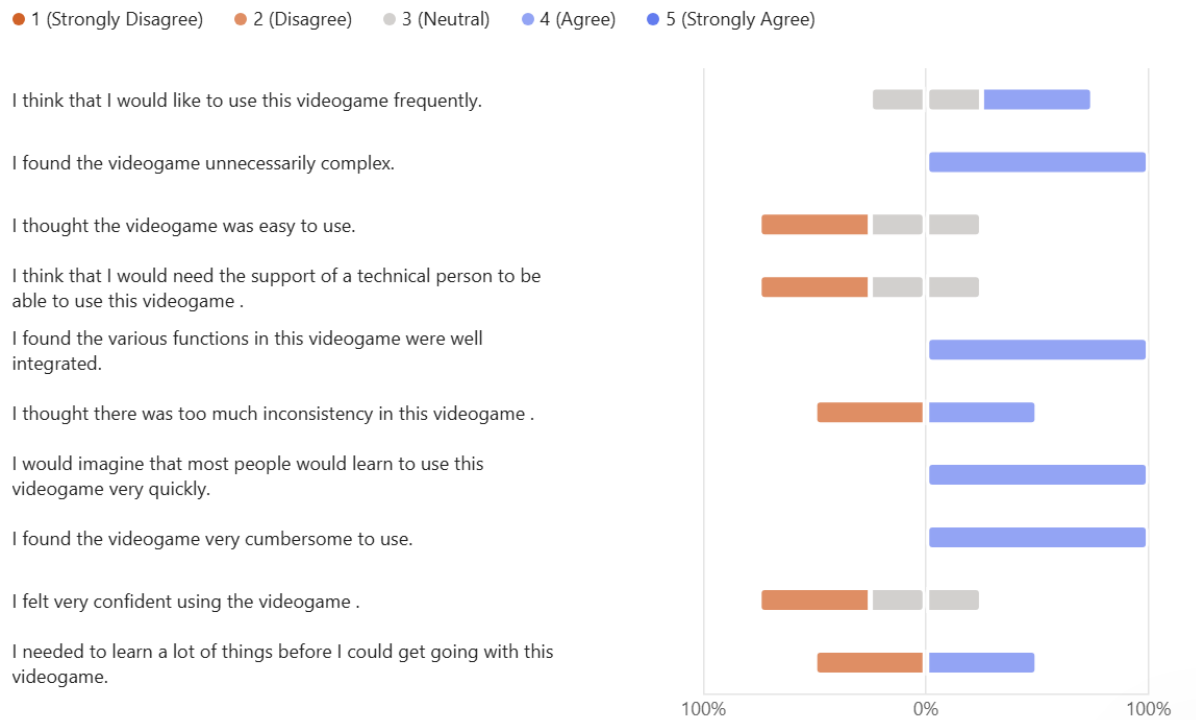


Figure 5.7 - System Usability Score Questionnaire

5.3. Discussion

The increase in precision could suggest that the CNG induced an improvement in positive predictions. However, an increase of 0.02 is negligible, and could be due to participants getting used to the landmark recognition task and undergoing the learning effect. It also could be random variation. The precision for both the pre-game and post-game tasks are very high, suggesting that participants were successful in identifying landmarks that were actually present on the route.

The recall for pre-game and post-game tasks was significantly lower than the precision. This suggests that participants rarely identified landmarks that were not in the route, but often did not select all of the landmarks present on the route. The increase of 0.045 from pre-game to post-game could suggest that the CNG influenced navigation by familiarising the participants with the landmarks on the routes. However, similarly to the precision, the small increase is not conclusive. It could be due to participants undergoing the learning effect. This effect is a large limitation of this study. This is because the routes, although different from pre-game to post-game, contain many of the same landmarks. Therefore, participants will start to recognise and remember landmarks simply from exposure from the study and use of Google Maps, not necessarily from the videogame.

The F1 Score increase of 0.053 comes from the precision and recall, and therefore may have come about for the same reasons.

The time decrease from pre-game to post-game tasks by 3 seconds could also be due to the learning effect, as the participants may simply have got faster at navigating between Google Maps, the landmark photos and the routes. They also may have gained a familiarity from having completed the pre-game tasks, as mentioned previously.

LICA, InfoLab and the Underpass were the landmarks with the greatest false negative count. This refers to the number of times which participants incorrectly classified them as not on the route. This could suggest that these landmarks were not modelled well in the CNG.

In general, the metrics measures are inconclusive in suggesting that the CNG improved the navigational and landmark recognition of participants. This is due to inherent weaknesses in the design of the study, leading to a possible learning effect. However, participants did suggest that they found that the CNG improved their navigation skill, with a confidence score increasing from a mean of 2/5 to a mean of 3/5 after playing the CNG. Furthermore, participants all reported being less reliant on GPS tools such as Google Maps after playing the game. These perceived results are backed up by the “think aloud” narration that participants underwent when completing the tasks. The narration showed that participants were able to identify familiar landmarks in the post-game tasks. However, these effects can all also be attributed to the learning effect, which may have happened without the participants realising.

Furthermore, it is difficult to attribute the small improvements in precision, recall and task completion time to the CNG as the study did not include in-game metrics, such as how much of the map they explored. Including this would have provided explainability as to which parts of the CNG led to improvements, but without it, it is difficult to pinpoint what parts, if any, had an effect.

Another key limitation of the study is the lack of participants. The study was conducted with only two participants, an extremely small sample size, due to difficulty recruiting participants to undergo a time intensive study. This makes it difficult to generalise these results, as they are very unreliable. The recorded metrics are therefore mostly invalid, due to the small sample size. Future studies should include a much larger sample of participants to increase the reliability and validity of results.

Overall, due to the limitations of the study, it cannot be claimed that the CNG caused significant improvement in the navigational and landmark recognition ability of participants for the Lancaster University campus. Other key requirements of the CNG, evaluated in this study, were that it should be replay-able and should not cause confusion. In order to evaluate these requirements, a SUS score was calculated. The result of this was a score of 50. This is below the benchmark average value of 68. This suggests that the CNG has noticeable usability issues, and is not clear and replay-able. Although this part of the study also suffers from the low sample size and hence is unreliable, the SUS score of 50 is significantly enough below 68 to suggest that it represents a real problem. This is reflected by participant feedback. After completing all of the tasks and playing the CNG, participants were asked to state any improvements that they would suggest for the CNG in an open-ended questionnaire question. One key theme was that participants felt there was a lack of clear instructions on how to play the game, especially how to play the cooking minigame. This reflects the SUS score, further suggesting that the CNG would benefit from better usability. The CNG contained a brief help screen for the cooking minigame, however this clearly was not enough. In order to make the cooking minigame and other mechanics clearer, a fully fledged tutorial could be implemented, taking the player step

by step through the minigame. Another option would be to simplify the minigame itself, by removing some features or making it more visually clear.

Participants also felt that the map should be more realistic in order to better match the actual campus. The map was designed with a low poly style, with certain landmarks modelled in more detail. This feedback suggests that more focus should have been put on making the map more realistic. Some features of the game could have been sacrificed in order to allow for more time modelling the map.

Finally, participants also felt that the game should have been more focused around navigation, and had too much focus on the cooking minigame where you are not doing any navigation. The cooking minigame was designed to be a fast-paced game that would not detract from navigation. However, due to the lack of clarity and detailed tutorial, participants found themselves getting stuck on playing the minigame. This detracted from the navigation portion of the CNG. Other options would be to focus more on the quests.

6. Conclusions

6.1. Review of Aims

The first aim set out for this project was to design and implement a videogame modelled on the Lancaster University campus. This aim was partially met. A map was designed which matched the layout of the campus. However, only a select few landmarks were modelled in detail. As mentioned by in the evaluation, a more realistic, detailed representation of the map would be desired, as this simplification impacted the other aims of the project. The second aim of the project was to incorporate gameplay intended to promote repeated exploration of the virtual campus and repeated exposure to campus landmarks. This aim was also partially met. The CNG implemented several features which facilitated exploration and landmark exposure, namely the delivery system, open-world exploration, quests and more. However, due to the unclear nature of the cooking minigame (a core part of the game loop), this detracted from this aim. The final aim of the project was to conduct a user study to measure the impact of the CNG on navigational ability of the campus. This aim was met, but with limitations. A study was conducted, comparing performance on a navigation task (identifying landmarks from a route) before and after playing the CNG. This study was inconclusive in showing any improvements resulting from the CNG, due to the small improvements in accuracy which could be attributed to the learning effect. Furthermore, a small sample size means that the results were further inconclusive.

6.2. Revisions of Design and Implementation

The user study highlighted several issues with the core gameplay of the CNG. Future work could refine the videogame to address these issues. The key improvements would be to fully model the game map on the campus, with a more realistic art style, including all buildings. This would better link the spatial memory of the virtual campus to the real-world campus. Another improvement would be to add a more comprehensive tutorial to the cooking minigame to make it less confusing, or to scrap the minigame and focus on mechanics that directly induce exploration of the campus, such as the quest system. In general, more focus should be directed to creating a faithful representation of the campus, rather than introducing complicated game mechanics that do not induce exploration of the game.

6.3. Future Work

Future work could implement a similar videogame with a more realistic representation of the campus. More work could be done to guide the player through the game, such as introducing a storyline. A larger-scale study could be carried out in order to gain some conclusive results. The study also did not focus on the implicit learning idea, so future research could test the replay-ability of a videogame focused on being a campus, and another that is not.

6.4. Personal Reflections

This project was the largest one that I have worked on alone. This presented challenges in terms of time management and scheduling, both balancing tasks within the project, and with external tasks such as other studies and extra-curricular activities. I have definitely gained an appreciation for the importance of good planning for a project of this size. Formulating a videogame from scratch was also a very challenging but rewarding experience, as I have limited experience creating full-scale videogames. Using Godot for the first time as the game engine was also an exciting learning experience, as most of my experience was with Unity beforehand. In addition, I had to be wary of project creep and constantly remain focused on the

core requirements of the videogame, despite the temptations to develop an extra mechanic that I would consider fun. I have also learned how difficult creating an engaging videogame is. Coming into the project, I thought that I would have no problem creating a fun game on the subject of campus navigation. However, once I started designing and developing the game, I realised how difficult that would be. I have gained valuable knowledge for future videogames that I may go on to develop. 3D modelling also provided a valuable challenge. Having only dabbled in Blender before this project and then having to model an entire university campus was a large step up. Although I made many mistakes and the final design was far from perfect, I have learnt a lot from the process. I have also learnt the difficulties of running an unfunded, crowd-sourced user study, but have developed my skills nonetheless. Finally, I have also enjoyed the challenge of creating a large report from scratch. This report is by far the largest piece of writing that I have ever created, which has developed my analytical writing skills to no end. It was also very rewarding to conduct background research into this area of study, and be able to contribute a substantial project to the space. In conclusion, I am very appreciative for the opportunities that this Third Year Project has given me.

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Appendix A. Project Proposal

173 Campus Navigation Game - Project Proposal

1. Abstract

This paper proposes a project involving the design and development of a browser-based exploration game which will use principles of implicit learning to familiarise prospective students with the campus layout of Lancaster University. The primary goal of this project is to ensure that students who play the game will acquire spatial knowledge of the campus environment, as well as knowledge of some of the systems in place, while engaged in an enjoyable and replayable experience. Hence, it should not be in the forefront of their minds that this is indeed an educational tool. The consequence of this gain in spatial knowledge is that students should feel more comfortable on campus and hence have a better university experience. The game itself will be inspired by the aesthetics and mechanics of 2.5D life-simulation and role-playing games such as *Stardew Valley* and *Octopath Traveller* – utilising quests, NPC interactions and a gripping restaurant delivery game loop to encourage repeated engagement and exploration of the in-game space. This spatial memory should then translate to when they arrive on campus and are exploring the real-world space.

2. Background

Reber posits that implicit learning is knowledge “acquired largely independently of the subject’s awareness of either the process of acquisition or knowledge base ultimately acquired”. In this case, the process of acquisition is the game created by this project, and the knowledge base is the spatial memory of the campus layout. This suggests that prospective students can learn the campus layout without deliberately trying to do so – a goal that the proposed system will aim to achieve.

Studies also show that implicit learning is especially common concerning navigational tasks. Frensch and Runger discuss the human “ability to adapt to environmental constraints – to learn – in the absence of any knowledge about how the adaptation is achieved” (Frensch & R  nger, 2003). An everyday example of this would be adapting to your commute to work; over time you unconsciously remember the route. The same can be said for adapting to a route and your surroundings in an immersive videogame.

Using a videogame to learn the layout of a campus is a new idea, however it could be a solution to many concerns about the existing way that students learn their way around campus and learn the systems in place. Most universities have a week before lectures and classes start that incoming students can settle in and are supposed to learn their way around. At Lancaster University this is called a “Welcome Week” (Lancaster University, 2025). These weeks have increasingly come under scrutiny for not being enough to allow students to properly ‘settle in’. One article from Campus Mind cites 3 key concerns with “Welcome Weeks”. The first and most pertinent is that of information overload. Students do not just have to memorise the campus layout, but they are also bombarded with multitudes of information about societies, sports, events, among many other things. When students have only 1 week before their classes start to absorb all of this information and also learn the campus layout, among many other systems, they largely forget it (Campus Mind, 2025). This argument is supported by Ebbinghaus’s forgetting curve research. Though an incomplete study, the hypothesis presented suggests that humans forget 90% of new information within the first 7 days, and 50% within the first hour (Ebbinghaus, 1913).

One way to remedy the forgetting curve is through repetition. There is a wealth of academic literature that discusses how repetition is integral for memory and learning, such as articles from Hintzman and Block (Hintzman & Block, 1971). A videogame introduced to students in the summer before university, that got students to traverse around the campus map repeatedly would introduce this repetition and cause players to remember most of the campus information for when they arrive there in person. In addition, students are not going to visit every corner of the campus in their first week, as there is no reason to. This means that certain areas in the campus remain unexplored and unknown to students until they may have a lecture or other activity there, leading to confusion. A videogame would give students an organic reason to explore every area of the campus, through the use of quests and non-playable-characters. Therefore, not only are students more likely to remember the campus through the use of a videogame, but are more likely to remember more of the campus.

Ensuring that the learning of the campus is implicit and not making it obvious that that is the aim of the videogame is key. Utilising the concept of implicit learning will mean that students should be more inspired to play. However, even if students do understand the primary purpose of the videogame, research suggests that this may not be a problem. De Freitas posits that the effectiveness of even overtly educational videogames is “overwhelmingly positive”, and suggests that the research points to “significant improvements in game over traditional methods” (De Freitas, 2018).

3. The Proposed Project

3.1 Aims and Objectives

The aim of the project is to create a videogame to be played by prospective Lancaster University students in the summer before they arrive at the university.

- The videogame should induce implicit learning in the students, whereby they subconsciously, or almost subconsciously, learn the layout of the campus and some of the systems in place.
- The videogame should be entertaining and replayable.
- The videogame should include exploration based quests to encourage players to traverse the campus environment.
- The videogame should have a consistent visual style.

The proposed videogame shall be a 2.5D life sim game with the key game loop of owning a restaurant and having to make deliveries across the campus environment (which will be a town).

Once the player receives an order, they will enter their restaurant and complete a cooking minigame where you prepare a recipe under a time constraint. They will then have to deliver the food to the NPC. The speed and quality of this process affects your reputation, and the player will receive money for delivering the food. You can also increase your reputation and earn money by interacting with NPCs and completing side quests for them. These quests will also allow the player to unlock ingredients to create new, more profitable recipes. They can also discover new tools, which for example would allow them to harvest certain ingredients (i.e., needing a fishing rod to collect fish). Some quests may lead to players unlocking new areas – such as helping a villager repair a road.

Delivering more meals, and especially more expensive meals, will give the player more money, which they can use to trade with villagers or buy from the shop. From this, you could buy ingredients, materials and tools to improve your restaurant or delivery system (i.e., unlocking a vehicle). Through helping the NPCs, the player will learn more about the story of the village and how it was ravaged by a mysterious creature. You help the village recover from this by helping villagers around town. There will also be quirky backstories behind characters to keep the interactions interesting.

This idea was settled on after extensive summer research and experimentation. Firstly, reading upon the subject of implicit learning and how a videogame could support this was completed,

along with thinking about how best to achieve this aim of implicit learning of the campus. Then, the idea of developing a Stardew Valley esque life simulation game was explored. This type of game is extremely replayable, and the quests force you to explore the map. This would be perfect for this project's objectives. Therefore, further research into Stardew Valley and similar games was conducted. Other similar games included Octopath Traveller, from which I was inspired by the different take on 2.5D. The 2.5D in Octopath Traveller included more depth, such that terrain could be different heights (Nintendo, 2018). This seemed important to include in this project, as many of the key features of the Lancaster University campus involve this additional level of depth, such as the Bigforth hill and the Bowerham Tower.

Once the style and genre of game was explored, the project needed a key game loop idea. In Stardew Valley this is your farm, which you expand and improve over time (ConcernedApe, 2016). Initially, I had the idea that you would play as a student, allowing for dormitory customisation and other missions. However, upon discussion with my supervisor, this seemed too on the nose and would not work with the key principle of implicit learning. Therefore, I explored other options and settled on a restaurant game loop idea. This would involve the player running their own restaurant where they make dishes and deliver them across the campus.

Whilst the decisions on the idea were being made, there was also a decision about how the videogame would be developed. This was essentially just a choice between an advanced game engine such as Unity or Godot, which would then have to be exported to use on the web browser, or a straightforward game made with HTML and JS. This could be supported with a game framework such as Phaser. The latter method would be more efficient, however the former would allow for much more flexibility and power in the approach. Therefore Unity was chosen as the technology to be used, as complex interactions such as those with NPCs and quests would be more easily implemented.

With these decisions being made, some preliminary Unity experimentation was carried out, testing out a basic version of a 2.5D environment, NPC movement, inventory systems and questing systems among other features. These were mostly developed independently simply to familiarise myself with the platform and what would be required.

3.2. Methodology

Object Oriented Programming shall be used, with C# classes being used in tandem with the Unity environment. A Waterfall software development methodology will be used, as there is little ongoing user feedback.

A 2.5D Stardew Valley and Octopath Traveller inspired pixel art aesthetic will be used, both as the game should be aesthetically pleasing, and this is less time demanding than other artistic choices such as full 3D modelling.

The testing and evaluation should analyse whether the videogame has been successful in achieving its objectives. The main objective was to create a videogame that causes players to implicitly learn the campus layout. This would ideally be tested on prospective students to the university, however the window for this evaluation would be the summer of 2026, when the dissertation would already be submitted. As an alternative, the project will aim to target people due to attend offer holder days. As a last resort, it would have to be tested on people unfamiliar with the Lancaster University campus, or people who maybe have visited once or twice – which would simulate the experience of a prospective student. This could involve being told to navigate to a certain area in the game environment before and after playing the game for a week.

The exploration objective could be evaluated with in game analytics, such as tracking the player's paths during a playthrough, to see whether they are forced to explore the entire campus.

4. Programme of Work

Stages:

- Background Research – This took place over Summer 2025 and involved research into the key principles behind the project. These included implicit learning, educational games, welcome weeks and memory. It also involved thinking about ideas about how to make the game achieve the specification and its aims, thinking which was supplemented with experimentation with the chosen game engine.
- Game Design – This involves fleshing out the initial project idea into a more tangible, concrete idea and set of aims.
- Write Draft Dissertation – This will take place alongside the game development and user evaluation process, discussing the background, implementation process, evaluation process and conclusions made.
- Code Development – This will take place during Term 1 and the Christmas break, involving the implementation of the videogame.
- Artistic Design – This will take place alongside the code development, involving the design of textures, character sprites, animations and 3D models for the videogame.

- **Evaluation Design** – This will take place towards the end of the game development stage, consisting of designing a suitable evaluation to test the aims and objectives of the project. Ethics approval will also be sought.
- **Perform Evaluation** – This will take place at the start of term 2, hoping to target incoming offer holder day students. Evaluation of the videogame will take place, testing against the initial aims and objectives.
- **Analyse Evaluation** – This will take place after the evaluation, consisting of analysing the evaluation results to see if objectives were met and whether any changes need to be made.
- **Implement Changes from Evaluation** – This will involve implementing any necessary changes highlighted by the evaluation analysis.
- **Final Alterations** – This will be at the very end of the project, after the evaluation and supervisor comments, clearing up any final bugs and implementing any final tweaks before submission.

Shown below in Figure 1 is a Gantt Chart showing the proposed programme of work.

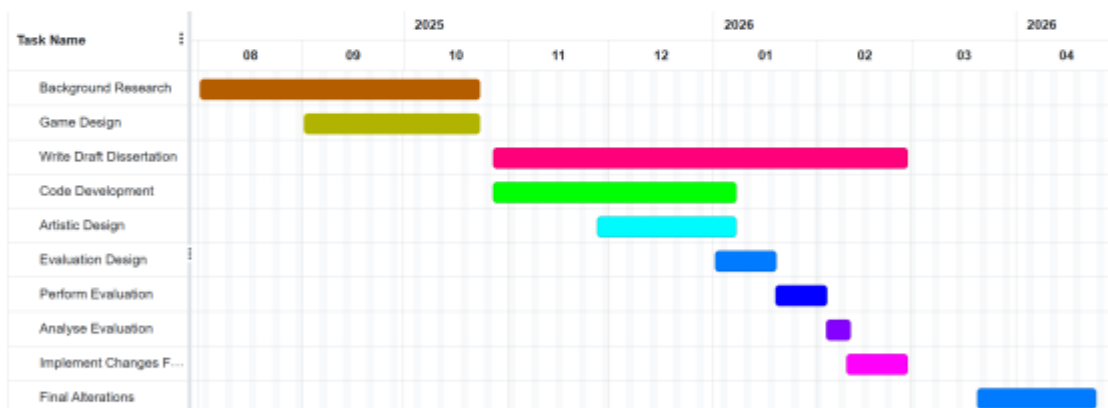


Figure 1 – TYP Gantt Chart

5. Resources Required

Access to Unity will be required for use as the game engine to create the videogame. A pixel art software such as pixilart.com will be needed to design the characters and the environment. A 3D design software such as Blender will be used to create the 3D models used in the environment. OpenStreetMap alongside Blender will be required to extract the campus map data and import it into the videogame. A web server will be needed to host the web game.

6. References

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Appendix B. Ethics Application

Dear Edward Vale,

Your UG Ethics application for the project titled "*TYP Campus Navigation Game*" has been reviewed by *Paul Dempster*.

Based on the information provided, **they have APPROVED the application.**

You can now start the study.

Additional Instructions:

- If the design of your study changes, you will need to submit another application.
- Please attach a copy of this email to the Appendix section of your TYP report
- Supervisor comments (if any): ""

Thank you.

Regards,

SCC Ethics Coordinator

--Application Contents--

Basic Information

Student Name: Edward Vale

Supervisor/Convenor: Paul Dempster

Course/Module Context

Is for TYP research (SCC.300)? Yes

Course name: Not Shown/NA

Module details: SCC.300 - TYP

Is group project? NA

Group Details

Size of group: Not shown/NA

Group member names:

Not shown/NA

Group member email ids:

Not shown/NA

Project Basic Information

Project Title: TYP Campus Navigation Game

Project Abstract:

This project involves the creation of a videogame, designed to be played by prospective Lancaster University students before they arrive at the campus for the first time. The videogame aims to induce implicit learning of the campus environment, in order to reduce the information overload that occurs in the first week of term.

Study Aims/Description:

This study aims to evaluate the effect of playing the aforementioned videogame on a person's navigation of the Lancaster University campus. It aims not to simply evaluate whether the navigational ability improves after playing the videogame, but instead how the methods of navigation are changed or enhanced, and whether they can identify campus landmarks to assist their navigation.

Does project involve a study/evaluation? Yes

Primary scope for participant type: Human participants (e.g. you will interview, observe, record people or their actions)

Human participants in secondary scope: Not Shown/NA

GDPR Compliance: Acknowledged. Confirm evidence in documentation.

Human Participants - 1

Basic information

Participant description:

Persons above the age of 18 who are not familiar with the Lancaster University campus. They must also have some level of computer proficiency so that they can play the videogame without extreme difficulty.

Participant numbers [min,max]: [2, 15]

Participant age range: ["18-25 years", "66+ years"]

Participant recruitment approach:

Participants will be recruited primarily via social media. For social media, an advert for the study will be published on Facebook and Instagram detailing the process and how to get involved. Snowball sampling will also be employed, so that previous participants can refer other suitable candidates to the study.

Human Participants - 2

Special category participants:

- My study does not require participants that definitely satisfy any of the above criteria

Checks for deception, trickery, undue stress and debrief:

- No. The study does not involve deception or covert methods (e.g. withholding essential information or intentional misinformation)
 - No. The study does not include activities that can cause stress or anxiety or involve physical contact.
-

Human Participants - 3

Consent and Privileged Data

Use of privileged data:

- No. My study does not collect data which falls in any of the above categories.

Ability to provide informed consent: Yes. The participants are able to provide informed consent when they participate in the study

Third-party consent: No, the study does not require such permission

Non-Human Participants

Non-human participants as primary scope of study? No

Non-human participants as secondary scope? No

Non-human participants details:

Risks

Indicated risk profile:

- No. I have carried out risk assessment and there is no risk to any individuals due to the nature of study or location of the study
-

Data Handling

What type of data is collected?

Information from the pre and post game tasks will be recorded. This will include the user's explanations of their navigation, as well as the images that they identified. Users will input their task results in a questionnaire, and the audio recordings from their think aloud task will be deleted but a text transcription will be stored. Data from the final questionnaire about the videogame will be recorded, including a SUS score, using Microsoft Forms.

Use of digital surveys: My study involves computer-based surveys or questionnaires and I confirm I will only use the approved tools.

Data storage strategy:

Transcriptions of recordings will be stored on the University OneDrive system, as well as any notes and other data. I will access these using a secure, password-protected PC. The questionnaire will be created using Microsoft Forms under the University account.

Anonymity and Confidentiality

Applicable situations:

- Participation in study is anonymous
- Plans for anonymisation ensure that participant cannot be identified indirectly
- Participants have been promised confidentiality
- Data processing will ensure participant confidentiality is maintained

Anonymity and confidentiality plans:

Personal information will not be recorded, and the responses and data collected will be fully anonymised. The voice-identifiable recordings will be deleted, and only text transcriptions stored. Only I (the researcher) will have access to the data, and it will all be stored on secure University OneDrive systems.

--End Contents--

Appendix C. Ethics Form

CONSENT FORM

Project Title: **CAMPUS NAVIGATION GAME**
Name of Researcher: **EDWARD VALE**
Name of Supervisor: **PAUL DEMPSTER**
Contact Email: **p.dempster@lancaster.ac.uk**

Please read the following carefully:

1. I confirm that I have read and understand the information sheet for the above study. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily. ☐
2. I understand that my participation is voluntary and that I am free to withdraw at any time, without giving any reason. If I withdraw within 3 weeks of commencement of the study my data will be removed. ☐
3. I understand that any information given by me may be used in future reports, academic articles, publications or presentations by the researcher/s, but my personal information will not be included and I will not be identifiable. ☐
4. I understand that my name/my organisation's name will not appear in any reports, articles or presentation without my consent. ☐
5. I understand that any interviews will be audio-recorded and transcribed and that data will be protected on encrypted devices and kept secure. ☐
6. I understand that data will be kept according to University guidelines for a minimum of 10 years after the end of the study. ☐
7. I agree to take part in the above study. ☐

Name of participant:	Date:	Signature:

I confirm that the participant was given an opportunity to ask questions about the study, and all the questions asked by the participant have been answered correctly and to the best of my ability. I confirm that the individual has not been coerced into giving consent, and the consent has been given freely and voluntarily.

Signature of Researcher/person taking the consent _____

Date _____ DD/MM/YYYY

One copy of this form will be given to the participant and the original kept in the files of the researcher at Lancaster University

Appendix D. Participant Information Sheet



Participant Information Sheet

Campus Navigation Game

For further information about how Lancaster University processes personal data for research purposes and your data rights please visit our webpage:

<https://www.lancaster.ac.uk/research/participate-in-research/data-protection-for-research-participants/>

I am a third-year Undergraduate student at Lancaster University, and I would like to invite you to take part in a research study about using videogames as a campus orientation technique, looking at how they effect navigation of the campus environment.

Please take time to read the following information carefully before you decide whether you wish to take part.

What is the study about?

This study aims to evaluate the effects of a playing a videogame developed for a Third Year Project. These effects will be evaluated in terms of the effect on navigation ability around the campus.

Why have I been invited?

I have approached you because I am interested in how people who are unfamiliar with the Lancaster University campus interact with the videogame, and how their navigational logic changes after playing the videogame.

I would be very grateful if you would agree to take part in this study.

What will I be asked to do if I take part?

If you decided to take part, this would involve the following:

You will firstly be asked to complete a handful of background questions. Then, you would have to complete a pregame activity. You would be presented with a series of images taken of landmarks around the Lancaster University campus. Then, a map of the campus with a route plotted will be presented. You would be required to identify which of the landmarks would be seen along the route shown. Use of Google Maps and MazeMap is permitted. You will repeat this for 4 routes, with 2 minutes allowed for each, but you can finish before then if you are satisfied you have identified all of the landmarks present on the route. You will also be asked to narrate your thought process during this task, recording these thoughts using a voice recording software. You will then record the time that it took for each route.

Then, you would be asked to play the videogame for 15 minutes, where you will complete certain quests inside of the videogame that require you to traverse the virtual campus environment.



After playing the videogame, you will be asked to complete the same navigation task as you completed before the videogame, with the same “think aloud” methodology, for 4 more routes.

Finally, you will have to fill out some short questions about the study in a questionnaire.

What are the possible benefits from taking part?

Taking part in this study will allow you to contribute to the exciting research area of using videogames as tools for learning.

It will also give you an opportunity to playtest a videogame, see inside that process, and contribute any improvements that you may see.

You will not be paid for participating in this study. There is unlikely to be any cost - financial or other - to you for participation in the study.

Do I have to take part?

No. It's completely up to you to decide whether or not you take part. Your participation is voluntary, and you are free to withdraw at any time, without giving any reason.

What if I change my mind?

As explained above, you are free to withdraw at any time and if you want to withdraw, I will extract any data you contributed to the study and destroy it. Data means the information, views, ideas, etc. that you and other participants will have shared with me. However, it is difficult and often impossible to take out data from one specific participant when this has already been anonymised or pooled together with other people's data. Therefore, you can only withdraw up to **3 weeks** after taking part in the study.

What are the possible disadvantages and risks of taking part?

The main disadvantage of taking part in this study is the time investment. You will have to commit around 55 minutes in total to complete the pre-game task, videogame playthrough and then the post-game task and questions.

Will my data be identifiable?

After the study, only I, the researcher conducting this study will have access to the data you share with me.

I will keep all personal information about you (e.g. your name and other information about you that can identify you) confidential, that is I will not share it with others. I will anonymise any audio recordings and hard copies of any data. This means that I remove any personal information. All reasonable steps will be taken to protect the anonymity of the participants involved in this project.

How will my data be stored?



Your data will be stored in encrypted files (that is no-one other than me, the researcher will be able to access them) and on password-protected computers.

I will keep data that can identify you separately from non-personal information (e.g. your views on a specific topic).

In accordance with University guidelines, I will keep the data securely for a minimum of ten years.

How will we use the information you have shared with us and what will happen to the results of the research study?

I will use the data you have shared with only in the following ways:

I will use it for academic purposes only – my Third Year Project. I may also present the results of my study at academic conferences.

When writing up the findings from this study, I would like to reproduce some of the views and ideas you shared with me. When doing so, I will only use anonymised quotes (e.g. from our interview with you), so that although I will use your exact words, you cannot be identified in our publications.

This study is being conducted based on the review and approval of the supervisor on behalf of the School of Computing and Communications, Lancaster University.

What if I have a question or concern?

If you have any queries or if you are unhappy with anything that happens concerning your participation in the study, please contact myself or Paul Dempster via the following details:

*Paul Dempster
School of Computing and Communications
InfoLab 21 Building, South Drive,
Lancaster University, Lancaster LA1 4WA, England
p.dempster@lancaster.ac.uk*

If you have any concerns or complaints that you wish to discuss with a person who is not directly involved in the research, you can also contact:

Professor Nigel Davies
*Head of School
School of Computing and Communications
InfoLab 21 Building, South Drive,
Lancaster University, Lancaster LA1 4WA, England
Email : n.a.davies@lancaster.ac.uk*

Thank you for considering your participation in this project.

Appendix E. Evaluation Walkthrough

User Evaluation Process

Thank you for agreeing to take part in this process.

The steps that this study involve are explained step by step below, but here is a brief summary.

You will first be asked to complete a short navigation task based on identifying landmarks from a map. You will have to record your thought process whilst completing the task, as well as recording your results. You will then be asked to play a videogame for a short time, and then complete another navigation task (again with narration) and answer some short questions.

Here are the detailed steps:

Step 1 – Complete Ethics Form

Before you complete this study, you need to read the **information sheet** and then **download** and fill out the consent form. Both documents are in the OneDrive folder. To download the consent form, click on the three dots next to name, and then select **download** from the dropdown.

Once you have read and filled out the consent form, please upload the completed consent form to this link:

https://livelancsac-my.sharepoint.com/:f/g/personal/valee_lancaster_ac_uk/IgAdSyeAwXLPQagDvB6SA8uNAX_ckLLB4WdxCHfTMQWMEyQ

I cannot use the results from the study unless you fill out and upload the consent form **first**.

Step 2 – Open Questionnaire

Once you have completed and sent over the consent form, please open the questionnaire. This is where you will input your activity results and answer some other short questions. The link is here:

[Campus Nav Game – Fill in form](#)

Please now complete the background questions in the questionnaire.

While you do this, it would be a good idea to download the videogame (**CampusNavGame.exe**) as it may take a few minutes to download depending on your internet connection.

Step 3 – Pregame Activity

Within the folder 'Landmarks', there are 11 images of landmarks around the campus.

Within the folder 'Routes', there are 8 routes across campus.

Your aim is to identify which of the 11 landmarks you would pass when navigating one of routes.

In the pregame stage, you will look at **routes 1 to 4**.

You will have **2 minutes for each route** to identify all of the landmarks that you would pass, but can finish before then if you are satisfied you have identified all of the landmarks present on the route.

Please also voice record yourself completing the tasks, indicating which route you are completing.

Please upload this voice recording to https://livelancsac-my.sharepoint.com/:f/g/personal/valee_lancaster_ac_uk/IgD2sLArQ1U7To91ZYKm_cVsAfHlt36lflo8q2hqeZTLksg

This file will be transcribed into text and the voice recording fully deleted so that any identifiable information will be removed and anonymity will be retained.

After each route, you will have to input the following into the questionnaire: the time it took you, and the landmarks you believe that you would pass.

You **may** use [Google Maps](#) and [MazeMap](#) to help you if necessary.

Once you have completed the routes, there are two multiple choice questions to answer in the questionnaire.

So:

Start a voice recording.

Then time yourself completing the task for routes 1 to 4, narrating your thought processes.

For each route, input your time and the landmarks you think you would pass into the questionnaire.

Finally, complete the multiple choice questions.

Step 4 – Game

Once you have completed the pregame activity, please open the videogame (**CampusNavGame.exe**) and play for **10-15 minutes**. The videogame should guide you through the basics, but feel free to explore for yourself.

Step 5 – Postgame Activity

Once you have played the game for the allotted time, you will now need to complete the postgame activity.

This takes the same format to the pregame activity, but now looking at **routes 5 to 8**.

You will have **2 minutes for each route** to identify all of the landmarks that you would pass, but can finish before then if you are satisfied you have identified all of the landmarks present on the route.

Please also voice record yourself completing the tasks, indicating which route you are completing.

Please upload this voice recording to https://livelancsac-my.sharepoint.com/:f:/g/personal/valee_lancaster_ac_uk/lgD2sLArQ1U7To91ZYKm_cVsAfHlt36lflo8q2hqeZTLksg

This file will be transcribed into text and the voice recording fully deleted so that any identifiable information will be removed and anonymity will be retained.

After each route, you will have to input the following into the questionnaire: the time it took you, and the landmarks you believe that you would pass.

You may use [Google Maps](#) and [MazeMap](#) to help you if necessary.

Once you have completed the routes, there are two multiple choice questions to answer in the questionnaire.

So:

Start a voice recording.

Then time yourself completing the task for routes 5 to 8, narrating your thought processes.

For each route, input your time and the landmarks you think you would pass into the questionnaire.

Finally, complete the multiple choice questions.

Step 6 – Postgame Questionnaire

Please now complete the postgame section of the questionnaire.

Step 7 – Finished

If you have completed steps 1-6, you are finished with the study!

Thank you so much for taking part.

If you know anyone else that would be willing to complete the study, then please either get them to email me (e.vale@lancaster.ac.uk) or just share this OneDrive folder with them.

Thanks again!

Appendix F. Evaluation Questionnaire

Campus Nav Game

* Required

Background

Please complete this section before completing any of the activities

1. How familiar are you with the Lancaster University Campus? *

- ☐ Very familiar
- ☐ Somewhat familiar
- ☐ Slightly familiar
- ☐ Not familiar at all

2. How would you rate your general navigation ability? *



Part 1 - Pregame

Please complete this section after completing the pregame activity

3. How long did it take you to complete the task for **Route 1?** *

4. What landmarks do you think you would pass in **Route 1?** *

☐ Landmark 1

☐ Landmark 2

☐ Landmark 3

☐ Landmark 4

☐ Landmark 5

☐ Landmark 6

☐ Landmark 7

☐ Landmark 8

☐ Landmark 9

☐ Landmark 10

☐ Landmark 11

5. How long did it take you to complete the task for **Route 2?** *

Campus Navigation Game

6. What landmarks do you think you would pass in **Route 2?** *

- ☐ Landmark 1
- ☐ Landmark 2
- ☐ Landmark 3
- ☐ Landmark 4
- ☐ Landmark 5
- ☐ Landmark 6
- ☐ Landmark 7
- ☐ Landmark 8
- ☐ Landmark 9
- ☐ Landmark 10
- ☐ Landmark 11

7. How long did it take you to complete the task for **Route 3?** *

8. What landmarks do you think you would pass in **Route 3?** *

- ☐ Landmark 1
- ☐ Landmark 2
- ☐ Landmark 3
- ☐ Landmark 4
- ☐ Landmark 5
- ☐ Landmark 6
- ☐ Landmark 7
- ☐ Landmark 8
- ☐ Landmark 9
- ☐ Landmark 10
- ☐ Landmark 11

Campus Navigation Game

9. How long did it take you to complete the task for **Route 4?** *

10. What landmarks do you think you would pass in **Route 4?** *

- ☐ Landmark 1
- ☐ Landmark 2
- ☐ Landmark 3
- ☐ Landmark 4
- ☐ Landmark 5
- ☐ Landmark 6
- ☐ Landmark 7
- ☐ Landmark 8
- ☐ Landmark 9
- ☐ Landmark 10
- ☐ Landmark 11

11. Did you use any of the tools during the pregame activity? *

- ☐ Google Maps
- ☐ MazeMap
- ☐ Both
- ☐ None

12. During the pregame activity, how confident were you in identifying landmarks around the campus? *



Part 2 - Postgame

Please complete this section after you have played the videogame and completed the postgame activity.

13. How long did it take you to complete the task for **Route 5?** *

14. What landmarks do you think you would pass in **Route 5?** *

- ☐ Landmark 1
- ☐ Landmark 2
- ☐ Landmark 3
- ☐ Landmark 4
- ☐ Landmark 5
- ☐ Landmark 6
- ☐ Landmark 7
- ☐ Landmark 8
- ☐ Landmark 9
- ☐ Landmark 10
- ☐ Landmark 11

15. How long did it take you to complete the task for **Route 6?** *

Campus Navigation Game

16. What landmarks do you think you would pass in **Route 6?** *

- ☐ Landmark 1
- ☐ Landmark 2
- ☐ Landmark 3
- ☐ Landmark 4
- ☐ Landmark 5
- ☐ Landmark 6
- ☐ Landmark 7
- ☐ Landmark 8
- ☐ Landmark 9
- ☐ Landmark 10
- ☐ Landmark 11

17. How long did it take you to complete the task for **Route 7?** *

18. What landmarks do you think you would pass in **Route 7?** *

- ☐ Landmark 1
- ☐ Landmark 2
- ☐ Landmark 3
- ☐ Landmark 4
- ☐ Landmark 5
- ☐ Landmark 6
- ☐ Landmark 7
- ☐ Landmark 8
- ☐ Landmark 9
- ☐ Landmark 10
- ☐ Landmark 11

Campus Navigation Game

19. How long did it take you to complete the task for **Route 8?** *

20. What landmarks do you think you would pass in **Route 8?** *

- ☐ Landmark 1
- ☐ Landmark 2
- ☐ Landmark 3
- ☐ Landmark 4
- ☐ Landmark 5
- ☐ Landmark 6
- ☐ Landmark 7
- ☐ Landmark 8
- ☐ Landmark 9
- ☐ Landmark 10
- ☐ Landmark 11

21. Did you use any of the tools during the postgame activity? *

- ☐ Google Maps
- ☐ MazeMap
- ☐ Both
- ☐ None

22. Compared to the pregame activity, did you rely less or more on external tools (Google Maps, MazeMap)? *

- ☐ Much more
- ☐ More
- ☐ Same
- ☐ Less
- ☐ Much less

Campus Navigation Game

23. During the postgame activity, how confident were you in identifying landmarks around the campus? *



24. Do you think that playing the videogame improved your ability to navigate the campus? *

- ☐ Yes, significantly
- ☐ Yes, slightly
- ☐ No chance
- ☐ Made it worse

25. If you think the videogame did improve your ability to navigate the campus, why do you think so? If not, why do you think it did not improve? *

Appendix G. Evaluation Think Aloud Task Transcripts

Participant 1

Route 1

00:01

OK, so I'm looking at the route now and I can see it goes past the Sports Centre, which I think is landmark 9.

00:16

And then as we carry on, I think it also goes past landmark 2 because I can see that the building is quite circular and then it goes, um.

00:31

We carry on up.

00:34

Along the map, just carry on and.

00:40

Round.

00:43

Through the trees.

00:48

Oh wait I've gone the wrong way.

00:53

Where else did they go?

00:58

Through here?

00:59

Ohh.

01:03

And then I think.

01:09

Ohh I think that's also a landmark.

01:11

1 maybe?

01:14

Campus Navigation Game

And then the route carries on.

01:17

Into the centre.

01:20

Which?

01:34

Help.

01:35

I can't get through here.

Route 2

00:00

OK, so now I'm looking at Route 2 and I can see that it goes all along the bottom.

00:08

I'm just walking along now and.

00:13

I can't see any landmarks so far.

00:17

Um.

00:20

I'm going down.

00:33

Umm.

00:51

Oh and I see

00:58

Oh and I see Landmark 7.

01:02

Along there because it's the same shape and colour.

01:08

Um.

01:16

Campus Navigation Game

And I also see Landmark 6

01:21

because it's also the same shape and colour.

01:25

I'm just carrying along.

01:29

Do do do.

01:35

Still haven't seen any more landmarks.

02:00

Now I'm at a roundabout.

02:06

Oh damn it.

Route 3

00:00

OK, so now I'm looking at Route 3.

00:04

Um, and I think I'm going to pass some tennis courts, which is landmark.

00:13

8.

00:15

Um.

00:17

Then I'm carrying on down the road.

00:25

Uh.

00:28

And.

00:45

I think I also go past.

01:02

Campus Navigation Game

Landmark 10.

01:05

Because it's going underneath.

01:09

The.

01:11

Underpass.

01:14

Which is going underground and then I also think that I go past.

01:29

Umm.

01:50

That's it I think.

Route 4

00:01

OK, so now I'm looking at Route 4, which firstly is going past um.

00:10

A big building.

00:11

Through some.

00:13

Um, fields which I don't think are in the landmarks.

00:18

And now I'm going past, um, some more buildings.

00:28

Which I also can't see.

00:35

And then I'm just carrying on down and going around.

00:57

Carrying one up

01:01

Campus Navigation Game

And I again see

01:02

Landmark.

01:05

7, going past that now.

01:11

And I'm just carrying on up.

01:18

And round.

01:27

And now I see Landmark

01:32

6.

01:44

Now I'm just going through.

01:46

I'm still going past Landmark 6 here.

01:50

And then I see Landmark

01:54

5.

01:58

Ohh, no.

01:59

And I also think I also see Landmark 4 and 3.

02:08

Boom.

Route 5

00:01

OK, now I'm doing route 5 and I'm just walking along the road now.

00:24

Campus Navigation Game

And I can see.

00:27

Um, landmark.

00:37

7.

00:45

And.

00:57

And I can see landmark.

01:02

11.

01:50

And I can also see Landmark

01.57

3.

Route 6

00:00

Now I'm doing Route 6 and I can see that I've gone past Landmark 2.

00:07

Um, and now I'm just carrying on along and.

00:19

Umm.

00:33

I'm also going past.

00:38

Landmark 1.

00:53

And then I'm just carrying along.

01:28

And I also think I'm going past Landmark 6

Campus Navigation Game

01:35

And yeah.

Route 7

00:00

Now doing Route 7 and I can see that I've gone past landmark.

00:09

Um.

00:16

9 which is the sports centre, and then I go down to the.

00:24

Umm.

00:29

Tennis courts, which is landmark.

00:35

Oh my gosh.

00:39

8 and then I also go past.

00:46

The.

00:52

Landmark.

00:58

Umm.

01:05

Ohh Landmark 10 which is the undercarriage and then I also go past Landmark 4.

01:19

As well.

Route 8

00:00

And now I'm doing.

Campus Navigation Game

00:03

Route 8.

00:07

Um, and I'm going past Landmark.

00:12

3.

00:16

Um, and.

00:25

Still going past Landmark 3.

00:30

Um.

00:34

And then I also go past Landmark.

00:43

10 which is the Underpass again

00:52

Umm.

00:57

And then just carrying on.

01:08

I also go past landmark.

01:17

I think.

01:19

No I don't.

01:22

No, I don't.

01:23

I'm in the wrong place.

01:28

Um.

Campus Navigation Game

01:30

I go past.

2:00

Landmark 11.

Participant 2

Route 1

00:01

OK, I'm looking at route one now which starts at the sports centre.

00:06

So what I'm going to do is just go into Google Maps, drop my pin down at sports centre.

00:13

So I'd imagine that this route must cover the sports field and potentially the tennis courts, although I can't see.

00:22

Them on here.

00:23

Well, I'm just going to go up the road now, towards the university where I'm turning left.

00:29

Luckily I actually recognise this road so I knew that I had to turn left to go up the hill.

00:35

I haven't seen any of the other buildings.

00:39

But just going to have a quick look back at the route, it looks like when I get to the roundabout I then have to turn left

00:49

Gone the wrong way I think

00:52

Yeah, I don't know why.

00:54

So not south drive, I need to go along.

00:57

North Drive down here.

01:01

Campus Navigation Game

Can't see anything that Oh yes, I've just gone past what that interesting shaped mosque building is.

01:08

So that's definitely on the route.

01:10

Just gonna have another quick look at the route now to see where I go.

01:16

I think I just keep going along here.

01:20

Past here, can't see anything else.

01:23

Going on North Drive.

01:28

I think I've gotta keep going until I sort of come to.

01:31

Another turning that goes down on the right there, but I can't really work out where that is unless it's down there, lets have a little look

01:44

Can't quite work out where you gotta turn.

01:52

I think I might have gone a bit too far around because.

01:56

Oh no, it does go quite round the corner, but I've.

01:59

I think I might have gone round too many corners now

02:02

And I've run out of time, so I've only got those two landmarks, I think, for route one.

Route 2

00:01

Alright, I'm now having a quick look at route um, two.

00:05

It starts at a roundabout goes all the way down to South Drive and ends up at Pendle College and then goes a little bit further round.

00:14

So I'm going into Google Maps now, um, and I can see that big.

00:20

Roundabout on South thing by Bailrigg motors.

00:23

I'm dropping down there, turning around so that I'm located going down the right way down, I can see a halls which blue and orange halls, which I think was one of the things.

00:36

I think that was number six it

00:40

Shows on the map that I've got to keep going down, um.

00:47

All the way down along S Drive, so I'm going to keep going.

00:52

There's an interesting orange and green building there, but I don't think that was on any of the landmarks.

01:00

Um, keep going along South Drive.

01:02

I've now got to another roundabout.

01:04

Sort of thing

01:05

So I'm just going to have a quick look back at the map.

01:12

Which I think is right, must be right next to Pendle College, but I don't quite know.

01:17

I'm obviously not as good at navigating as I thought.

01:21

Yeah, I think that must be Pendle College there on the right.

01:27

Campus Navigation Game

I can't see any signs that tell me that or anything.

01:31

But when I go back to the map.

01:36

It looks like I go round to the right and then off to the left a little bit, so I'm just going to have a quick look down there.

01:46

I'm now going down Alexandra Park, which I think is right.

01:50

Yes, I've just checked the map and that is correct.

01:54

So, um going down here.

02:00

I'm just going to check the time.

02:04

I've gone too long now.

Route 3

00:00

OK, I'm going to have a look at route 3 now, which goes down from near the Sports Centre all the way up through to the underpass.

00:09

So, um, just going on to the Google Maps.

00:12

It's just off of Bigforth drive from the map, which I can see, I can see the Sports Centre on.

00:20

The left, um, the tennis courts are on the right, which is 8, so I've got that one.

00:27

I'm going to go up Bigforth drive, but I don't know how I get across there.

00:36

So what I'm going to do is just come out of this view and then go back up to South Drive

00:44

Campus Navigation Game

Um, which I've just got to check the map because you end up on South Drive near Farrier Avenue

00:52

So where is that?

00:53

There's Farrier Avenue

00:54

And there's South Drive, so I'm going on to there.

00:59

So that's quite close to the.

01:04

I don't know what that building is there, the medical centre, I think.

01:09

So then I have to.

01:12

Sort of go back on myself there in the other direction.

01:18

And then I think I turn down there, do I?

01:25

Have another look at the

01:26

No, I've got to get back up to the roundabout and then.

01:31

I haven't seen any other oh I've seen that.

01:34

I don't know if that's one of the landmarks that funny shaped building, but there's definitely the underpass here.

01:41

So I'm just going through under there and that so I've seen a couple of landmarks there.

01:48

Um.

01:50

But I don't think I've seen more than.

Campus Navigation Game

01:54

2.

01:57

So that was the underpass, which is um.

02:03

Landmark 10.

Route 4

00:00

OK I'm onto route 4 now which seems to go up Green Lane.

00:07

Uh, then it goes past the Grizedale Pendle College, um.

00:14

Then it goes up South Drive.

00:16

I'm going to look at the map properly this time because.

00:21

It seems to go off into the South spine which, and into the centre of the campus, which I think is where then you will have gone past halls.

00:32

I think you'll go to the Union which is 3 and I think you've gone past the orange and blue halls and.

00:42

Which are number five I think.

00:48

And possibly number six, um, I'm just going to have a quick look at this time.

00:53

I'm not going to do the whole route.

00:56

I'm just going to drop in at near Fylde.

01:00

Residences, I think.

Campus Navigation Game

01:08

That's the right way that's the orange and blue one so yes.

01:13

So that's definitely on there, which was six um.

01:26

On Furness now, which I think might have gone wrong actually, oh no going past Furness College

01:38

And it's a bit of a dead end there

01:42

Need to get on to the South spine?

01:46

Which is difficult to do.

01:47

I think I'm at a dead end now.

01:57

Um, let me just go back out and then what I'm going to do is, is just try and.

02:04

Go back in on the centre, um.

02:09

Where I should end up

02:11

So going up the South spine which I could see there.

02:16

Um.

02:18

Then actually going right out of that.

02:21

Yeah.

02:21

And that takes me up to the union.

02:29

Campus Navigation Game

Which is at the top there, I believe I can see that now.

02:33

So that's the Union, which is landmark number three.

02:36

Let me just have a quick look at the map.

02:40

I think that might be.

02:43

Where it finishes, um, yeah, so I think I've done that one.

Route 5

00:00

OK, I'm looking at Route 5 now, which goes up along, um, Alexander Park Drive

00:06

and then down to the South Drive.

00:08

I'm going to do that bit first.

00:11

I've just gone up there, looking on the Google Maps, going along South Drive

00:16

There's some halls there.

00:19

I don't think I've seen any of those.

00:22

On the

00:24

Things.

00:25

What about that?

00:27

Green Science building, I think I've got there on the right, which is 7 um, carrying on along.

00:36

South drive and then I do need to turn down South spine.

Campus Navigation Game

00:45

Which I can't quite work out where that is.

00:55

I think it might be just after that.

01:01

Green building.

01:09

Just down there

01:15

Ohh yes.

01:16

And that's taking us past those halls.

01:19

Um, which is um.

01:22

Number 11 going past there.

01:31

I'm just following that down.

01:39

That's the South spine I believe, which is then going to take me up to.

01:52

The student union.

01:56

Let me just double check the map.

02:01

I think I might have gone too far.

02:06

Down the South spine because I should have turned left.

02:14

Down the Farrier Avenue

02:28

So I think I just need to come back out of there.

Campus Navigation Game

02:32

Quickly and um.

02:37

Have a look at where that Farrier Avenue

02:40

is which is down there.

02:48

That takes, you know, I can't see any other landmarks and I think I've run out of time now.

Route 6

00:00

OK, I'm now on to number 6 So this one goes down South Drive again and pass that roundabout.

00:07

So I'm just going to.

00:10

Go and have a look there

00:12

So I think that must therefore go past the um.

00:17

Blue and orange halls, which I think was number six, and it also definitely ends up by the mosque as I could see that circle of buildings.

00:27

So that's number two.

00:31

So if we just go along there

00:39

What I'm going to do actually, because I've gone around that roundabout, just keep going and get onto North Drive.

00:46

I don't know whether those halls, those white halls are on the list.

00:53

Campus Navigation Game

And then it bends around to the left.

00:56

I can see that on the map down following down North drive.

01:00

But before I get down there I need to cut down through there past that white building.

01:07

Is that one on the list?

01:10

Not sure what that was, Um.

01:13

One might possibly be um.

01:19

I'm just gonna double check the map again because this is where it gets a little bit confusing, so I think I might come out of the looking at it that way and go back into.

01:35

looking from above.

01:37

That seems to sometimes be easier to look at things.

01:42

So yes, it looks like you go down there past the Great Hall Square.

01:50

And then back to the, I presume you end up at the mosque.

01:53

I'm just going to have a quick look around there.

02:01

I'm not sure where I am now.

02:04

No.

02:10

Somewhere near Bowland College but I've run out of time again.

Route 7

00:00

And now on to Route 7.

00:02

I'm going to try a slightly different approach here because I know that, um, it's starting at the Sports Centre, UM, so I can definitely see that it will cover the sports field, which is number 9.

00:16

It then goes around the tennis courts, which is number 8.

00:21

I can also see that goes up big 4th drive and then actually goes past that roundabout which I'm now familiar with and it goes through the underpass, which is number 10 um.

00:35

So then it goes down the South spine a little bit and ends up at Fylde college.

00:41

So I'm actually going to just go straight to there on Google Maps, um.

00:46

Rather than.

00:48

The other way, um.

00:52

I just need to try and find that on the map.

00:57

Which I could just Google it, couldn't I?

01:02

Fylde College.

01:06

And that showed me there.

01:11

Yeah.

01:11

And so that's looking, Yeah, I can see that there now.

01:14

Campus Navigation Game

Um, and I think I would have gone past the Spar and the Costa and everything to get to that.

01:20

I'm just going to have a quick look on the roof again.

01:24

But yes, it would have gone past that circular bit, which I looked at before and I could see that.

01:30

Was that so, um, I'm just going to check that on the.

01:37

On the Google Maps um.

01:40

Costa.

01:44

Lancaster University.

01:46

Bailrigg Lancaster.

01:49

Because it's in that circle a bit, which makes it yeah.

01:53

So I think I've found quite ohh, yes, Costa, I've just found that on there.

01:58

So that's good.

01:59

Um.

02:00

And I've done that in time as well.

Route 8

00:00

I am on 8 now so I'm just gonna have a quick look at the route which takes us down along South spine that goes past um.

00:14

11, I believe the halls there, then as you're going along the South spine, I can see from the previous like going around.

00:23

So that will take us past the Student Union, which is number 3 and then it goes along the north spine and that's where I start to get a bit.

00:34

Disorientated because I'm not as familiar up there, so I'm going to have a look at the north spine on Google Maps.

00:44

Bailrigg.

00:47

Um, drop down onto there where it sort of goes off.

00:52

This is where it gets a bit confusing though, because.

01:04

I can't quite work out where I'm going.

01:06

I'm near.

01:11

No, that's all.

01:14

Lost my orientation now

01:15

What I am facing?

01:22

So it goes up to the Arts Centre, I think, but.

01:27

Yeah, it goes.

01:28

Theres the County College up there

01:33

And then you veer off and then up towards the top of North Drive, but I can't think what other.

01:39

Campus Navigation Game

Landmarks are going to be on route that way.

01:42

Um, we go back into the Google.

01:46

And just see what I can see along here.

01:51

Think it might take us past

01:55

Bowland.

02:00

Might be one of those other halls, but I've run out of time now.